The Magnetization Process of Certain Red Beds: Vector Analysis of Chemical and Thermal Results†

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A vector analysis of chemical leaching and thermal demagnetization results of 179 specimens of the Hopewell Group shows that three magnetizations were acquired in a nearly parallel direction but at different times as indicated by polarity. Field evidence indicates that magnetization occurred within 35 m.y. of deposition. It is suggested that these red beds were magnetized in three stages during a magnetization process which began at the time of deposition; hence producing one detrital (DRM) and two chemical (CRM_A and CRM_B) remanent magnetizations in that order. The two CRMs can be successively removed chemically to uncover the DRM. The CRM_B is demagnetized by thermal treatment at 600–674 °C, but, the DRM and the CRM_A cannot be separated thermally. This accounts for apparently aberrant directions obtained after thermal demagnetization at 674 °C, because the resultant vector of the remaining (and sometimes oppositely directed) DRM and CRM_A is not directed along the direction of the magnetizing field. However, by vector analysis of the combined chemical and thermal results, the directions of the three magnetizations can be determined and an accurate field direction (174°, +15°; α_{95}: 3°; pole: 36° N, 123° E) is thus obtained. More of the information contained in the rocks can also be retrieved from the within-specimen magnetization observed by cutting specimens at some stage during chemical or thermal treatment. For example, the results indicate the following about field reversals: Several of them occurred at the Pennsylvanian–Mississippian boundary; the intensity of the field may remain constant at the beginning of a reversal; reversals may be rapid, and, in some instances, of short duration, leading to the suggestion that, at reversal time, the field may be subjected to an oscillatory motion before stabilizing in the same or opposite polarity.

Une analyse vectorielle de résultats obtenus par immersion dans l'acide hydrochlorique et par désaimantation thermique de 179 échantillons du groupe Hopewell indique que ces grès rouges acquirent trois aimantations presque parallèles. Quoique ces aimantations ne furent pas acquises en même temps, puisque les polarités sont différentes, un plissement de terrain témoigne qu'elles eurent lieu en deçà de 35 m.a. du dépôt de ces sédiments. Nous suggérons donc que ces grès rouges furent aimantés en trois étapes pendant un procédé d'aimantation qui débuta au dépôt et qui produisit une aimantation détritique (DRM) et deux aimantations cristallines (CRM_A et CRM_B) successivement et selon cet ordre. Par immersion dans l'acide hydrochlorique, il est possible de détruire successivement les deux aimantations cristallines et ainsi dégager l'aimantation détritique. Le traitement themique permet de désaimanter la CRM_B mais, ce qui est très important, ne peut séparer la DRM et la CRM_A signifiant donc qu'un résultat obtenu par désaimantation thermique (même aussi haut que 674 °C) n'indique pas nécessairement la direction du champ d'aimantation. Cependant, au moyen d'une analyse vectorielle de résultats obtenus par traitements chimique et thermique, on peut définir la direction de chacune de trois aimantations et de là déterminer avec précision la direction (174°, +15°; α_{95} = 3°; pôle = 36° N, 123° E) du champ magnétique de l'époque. L'agrégation de trois aimantations dans le même échantillon permet d'étudier certains aspects du champ magnétique dans le passé. Ainsi, au sujet des renversements de polarité du champ magnétique, il apparaît que plusieurs eurent lieu vers la limite du Pennsylvanien–Mississippien; nous observons que l'intensité du champ peut demeurer constant au début d'un renversement; de plus, les résultats indiquent que les renversements peuvent être rapide et quelquefois de durée courte ce qui nous conduit à suggérer que le champ peut parfois être sujet à un mouvement oscillatoire lors d'un renversement.

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Introduction

Previous thermal demagnetization studies of samples from the Maringouin (Roy and Robertson 1968), the Shepody and the Enragé (Roy and Park 1969) Formations showed that, at many sites, the within-site dispersions, after thermal cleaning, were very large. This was attributed to the presence of two thermally stable magnetizations acquired, one before and one after, a field reversal, and it was concluded that much of the information contained in those red beds could not be obtained from thermal experiments alone. Preliminary thermal demagnetization and chemical leaching experiments conducted on six specimens of the Maringouin and Shepody Formations (Roy and Park 1972) showed that three magnetizations were acquired in the early history of those beds; that is, before folding which occurred within 35 m.y. of deposition. Because (*1*) two of the magnetizations could be successively removed by chemical leaching while the third remained unaffected, and (*2*) the magnetizations had different polarities, it was suggested that one detrital (DRM) and two chemical (CRM_A and CRM_B) remanent magnetizations were acquired at different times and over a period of time long enough to span field reversals. If so, then red beds are a valuable source of information for the study of certain features (*e.g.*, field reversals) of the earth's field in the past. However, the problem is to extract that information, and the preliminary experiments indicate that in order to do so a special procedure is needed. Those results show that only chemical leaching can isolate one (the DRM) of the three magnetizations. The DRM and the CRM_A are both thermally stable and the magnetization remaining after thermal demagnetization at 674 °C is composed of those two magnetizations. However the CRM_B can be separated from the DRM and CRM_A by thermal demagnetization at temperatures of 600 °C or higher, and sometimes by limited chemical leaching. The two cleaning techniques complement each other, and it appears that an analysis of combined thermal and chemical results might provide much of the information contained in red beds.

In this article, we describe the procedure used to separate, by vector analysis of thermal and chemical results, the three (DRM, CRM_A, and CRM_B) magnetizations contained in red-bed samples of a new collection of Carboniferous rocks. The within-specimen magnetization of some samples is also studied; the procedure used for that study is described in the section *Sites VIII, IX, X, and XI.*

Procedure

A section of the Maringouin and Shepody Formations on the south limb of the Maringouin anticline and on the east coast (295.5° E. 45.7° N) of the Maringouin Peninsula in New Brunswick, Canada, was sampled. The stratigraphic positions of these formations are given in Table 1. Core samples (111) were collected at 17 sites, and where possible, two specimens (labelled *A* and *B*) were cut from each, yielding a total collection of 179 specimens. The relative stratigraphic levels of the specimens and sites are given in the Appendix. Generally, one specimen from each core was treated thermally and the other chemically; in a few instances, both were subjected to the same treatment as indicated in Tables 2, 3, and 4. The chemical treatment consisted of immersing in hydrochloric acid for a given time, washing in water, measuring and reimmersing in fresh acid. The leaching rate was optimized by drilling a one centimeter axial hole (Park 1970).

It appears that three (DRM, CRM_A, and CRM_B) magnetizations are present in all specimens. The direction of each magnetization can usually be determined by vector analysis of the NRM (natural remanent magnetization), the thermally cleaned, and the chemically cleaned results. However, the thermal stability spectrum of each magnetization and the solubility (in acid) spectrum of the grains carrying the different magnetizations vary greatly amongst specimens so that a single cleaning temperature (for thermal) or immersion time (for chemical) is not usually sufficient to determine the points at which the magnetizations are separated. Therefore all specimens subjected to thermal treatment were demagnetized in a non-magnetic oven (Roy *et al.* 1972) at successive temperatures of 400, 560, 600, 650, 666, and 674 °C and then cooled in a quasi-zero ($<2\gamma$) field (Roy *et al.* 1969). A few of these specimens were also demagnetized at temperatures

TABLE 1. Age information relating to certain Paleozoic rock formations (from Roy 1969)

Formation or group	Probable age of formation	Age of Magnetization		Polarity	Reference
		Prior to or	Determined by		
P.E.I. sediments	Early Permian or late Pennsylvanian	Permian	Divergence from Triassic or later directions	+	Roy (1966)
Pictou Gp. (Tormentine Fm.)	Late Pennsylvanian	Permian	"	+	Roy (1966)
Bonaventure Fm.	Pennsylvanian (?)	Permian	"	+	Roy (1966)
Hurley Creek Fm.	Late Pennsylvanian	Permian	"	+	Roy *et al.* (1968)
Cumberland Gp.	Middle Pennsylvanian	Pictou	fold test	+	Roy (1969)
Enrage Fm.	Early Pennsylvanian	Cumberland Gp.	fold test	+	Roy and Park (1969)
Shepody Fm.	Late Mississippian	Cumberland Gp.	fold test	M*	Roy and Park (1969)
Maringouin Fm.	Late Mississippian	Cumberland Gp.	fold test	M*‡	Roy and Robertson (1968)
Perry Fm.	Late Devonian	Early Pennsylvanian	folding effect	+	Robertson *et al.* (1968)
Bloomsburg Fm.	Late Silurian	Appalachian folding	fold test	−(M?)*	Roy *et al.* (1967)

*Mixed polarity
‡Revised

of 100, 300, 625, and 682 °C. Except where otherwise indicated the chemically treated specimens were immersed in acid for 4545–4745 hr. During that interval, they were measured 20 to 25 times. Unless otherwise indicated, the acid strength was 8N for the first 1300 hr of leaching and 10N thereafter. Examples of different vector changes during thermal or chemical cleaning are given in the figures. The results obtained directly from the thermal and chemical experiments and which are used for the vector analysis, and the results obtained from the vector analysis are listed in Tables 2, 3, and 4. An astatic magnetometer with negative feedback (Roy *et al.* 1972) was used to measure directly the directions (declination and inclination) and intensity of magnetization.

The specimens from sites I to VII are fine-grained sandstones belonging to the Maringouin Formation; those from sites VIII to XVII are medium-grained sandstones belonging to the Shepody Formation. All directions quoted (and shown in the figures) are with respect to bedding.

Presentation

Since much varied information is obtained from different sites, the results are firstly presented according to the numerical order of the sites (or groups of sites). Together with this presentation, observations and findings applicable to, and conclusions drawn from, the results of that particular site are given. This section is concluded by a study of viscous magnetization to determine the reliability of the final results, and a study of polished sections to determine the carriers of the different magnetizations.

The next section presents a synthesis of the results, and the derived conclusions. It includes an evaluation of the results obtainable from the two cleaning techniques when used separately or together, with or without vector analysis; a proposed process of magnetization for red beds; and a discussion about the behaviour of the field during field reversals. A synopsis, reviewing the main points of the study, concludes the paper.

Results

Site I

The NRM is southerly directed (Table 2). After thermal demagnetization at progressively higher temperatures (20 to 600 °C), the direction is gradually displaced towards the north (Figs. 1*a* and 2*a*) and reaches an end-point at temperatures from 600 to 674 °C. In this latter range, the intensity sometimes remains approximately constant (Fig. 1*c*) and sometimes decreases slightly (Fig. 2*d*). With chemical leaching, the vector is also displaced

TABLE 2. Vector analysis of DRM, CRM_A, and CRM_B of sites I–VII

Specimen	treatment	NRM Mi	NRM D°,I°	RV: DRM+CRM_A THERMAL Mi	RV: DRM+CRM_A THERMAL D°,I°	RV: DRM+CRM_A CHEMICAL Mi	RV: DRM+CRM_A CHEMICAL D°,I°	CRM_A THERMAL Mi	CRM_A THERMAL D°,I°	CRM_A CHEMICAL Mi	CRM_A CHEMICAL D°,I°	CRM_B Mi	CRM_B D°,I°	DRM Mi	DRM D°,I°
							SITE I								
1A	th	5.91	187,+11	1.72	349,−23			2.62	352,−19			7.54	183,+14		
1B	ch	4.52	195,−07			3.00	344,−26			3.87	348,−22	6.98	183,+06	0.92	178,+10
2A	th	4.96	183,+05	2.46	339,−17			3.26	345,−15			7.25	175,+09		
2B	ch	5.82	173,+12			2.60	338,−23			3.38	344,−19	8.33	169,+15	0.85	181,+07
3	th	4.91	172,−03	2.42	342,−37			3.32	347,−29			6.92	169,+10		
4A*	ch	4.06	164,−43											0.96	180,+05
4B*	ch	4.55	183,−19											0.89	185,+06
5A	th	4.89	182,−02	2.60	340,−40			3.17	344,−32			6.94	176,+12		
5B	ch	5.52	182,−13			2.44	331,−41			2.98	338,−32	7.03	174,+03	0.72	174,−01
6A	th	4.26	179,−30	3.48	336,−44			4.08	342,−37			6.08	170,+03		
6B	ch	4.52	187,−36			3.32	347,−49			3.91	351,−41	5.75	180,−01	0.81	181,+03
7A	th	5.55	178,−33	3.37	331,−45			4.49	339,−35			6.89	169,−05		
7B	ch	3.84	184,−21			3.07	348,−56			4.12	350,−42	5.38	179,+13	1.36	173,+09
MEAN		*4.89*	*181,−14*	*2.68*	*340,−34*	*2.89*	*341,−39*	*3.48*	*345,−28*	*3.65*	*346,−31*	*6.83*	*175,+07*	*0.93*	*179,+06*
(*k*)			(17)		(41)		(28)		(69)		(50)		(91)		(208)
							SITE II								
1A	th	10.5	185,+14	2.65	195,+15			1.48	209,+18			7.90	182,+14		
1B	ch	10.6	193,+07											1.27	179,+10
2	ch	9.0	184,+26											0.82	169,+22
3A	th	12.0	180,+01	3.90	189,−01			3.40	192,−04			8.17	176,+02		
3B	ch	11.1	178,+12											0.57	170,+17
4	th	5.7	184,+11	2.60	198,−02			1.93	210,−10			3.35	172,+21		
5A	ch	10.5	179,+12											0.72	160,+06
5B	th	10.5	181,+11	2.52	188,+14			1.92	199,+16			8.01	179,+10		
6A	ch	14.6	180,+07											1.07	179,+18
6B	th	12.1	182,+11	2.50	198,+13			1.53	211,+09			9.72	178,+10		
7A	th	15.4	181,+10	3.96	197,+14			3.19	204,+13			11.65	176,+09		
7B	th	11.6	179,+12	1.85	185,+23			1.03	199,+27			9.80	178,+10		
8	ch	10.6	177,+13											0.95	170,+17
9	th	10.5	183,+08	3.08	187,+07			2.27	193,+03			7.43	181,+08		
10	ch	9.5	174,+20											0.79	165,+31
MEAN		*10.9*	*181,+12*	*2.88*	*192,+10*			*2.09*	*202,+09*			*8.25*	*178,+11*	*0.88*	*170,+17*
(*k*)			(130)		(66)				(32)				(171)		(60)

TABLE 2 (continued)

Specimen	treatment	NRM Mi	NRM D°,I°	RV: DRM+CRM$_A$ THERMAL Mi	THERMAL D°,I°	RV: DRM+CRM$_A$ CHEMICAL Mi	CHEMICAL D°,I°	CRM$_A$ THERMAL Mi	CRM$_A$ THERMAL D°,I°	CRM$_A$ CHEMICAL Mi	CRM$_A$ CHEMICAL D°,I°	CRM$_B$ Mi	CRM$_B$ D°,I°	DRM Mi	DRM D°,I°
								SITE III							
1	th	14.6	170,+14	2.76	176,+09			1.51	177,+08			11.87	169,+15		
2	ch	15.6	175,+24											1.36	162,+17
3	th	16.5	177,+10	4.91	179,+09			3.66	179,+09			11.60	176,+10		
4	th	18.2	182,+04	2.58	194,+05			1.46	210,+00			15.69	180,+05		
5	ch	15.4	172,+13											1.47	170,+14
6	ch	15.2	174,+17											0.69	191,+04
7	ch	15.6	170,+20											1.16	181,+03
8A	th	16.0	174,+20	3.65	180,+18			2.15	186,+24			12.27	172,+21		
8B	ch	16.3	173,+18											1.55	172,+10
9	th	15.9	172,+20	2.76	183,+20			1.56	190,+27			13.19	170,+20		
MEAN		*15.9*	*174,+16*	*3.33*	*182,+12*			*2.37*	*188,+14*			*12.92*	*173,+14*	*1.25*	*175,+10*
(*k*)			(142)		(74)				(22)				(100)		(42)
								SITE IV							
1	ch	2.0	176,+03			0.60	073,−34			1.98	358,−12	2.21	189,+11	1.87	163,+02
2	th	2.9	285,−50	6.00	340,−22			7.36	342,−19			4.75	179,+00		
3A	th	3.8	164,+00	2.17	357,−34			3.22	355,−24			5.70	168,+12		
3B	ch	10.1	158,+10			0.40	162,−09			0.78	359,−14	9.72	158,+11	1.15	173,+06
4	ch	3.0	151,−04			0.86	109,−32			1.21	015,−20	2.51	162,+06	1.39	164,−02
5	th	4.1	135,−30	5.64	010,−31			6.88	005,−26			7.51	167,+07		
6A	th	7.2	171,+11	3.06	343,−19			4.30	347,−17			10.22	169,+13		
6B	ch	4.3	170,+21			0.53	151,−08			0.87	012,−23	3.87	173,+25	1.27	176,+12
MEAN		*4.7*	*166,−05*	*4.22*	*352,−27*	*0.60*	*128,−25*	*5.44*	*352,−22*	*1.21*	*006,−17*	*5.81*	*171,+11*	*1.42*	*169,+05*
(*k*)			(4)		(32)		(4)		(63)		(68)		(46)		(85)
								SITE V							
1	th	11.0	180,+28	3.50	359,−11			4.31	359,−12			14.4	180,+24		
2	ch	9.7	183,+30			0.99	001,−02			1.76	354,−07	10.6	183,+27	0.80	165,+14
3	th	12.0	180,+25	2.90	009,−04			3.69	007,−06			14.7	182,+21		
4A	ch	9.1	190,+32			0.19	270,+42			0.93	338,−10	9.0	189,+31	0.92	167,+18
4B	th	9.6	190,+26	2.71	329,+09			3.52	333,+02			11.4	181,+19		
5	th	12.4	174,+18	3.25	341,+03			4.00	345,+01			15.4	171,+14		
6A*	ch	8.4	236,+32											1.04	178,+19
6B*	ch	6.5	191,+22											0.87	176,−02
7	th	10.4	202,+31	3.37	325,+20			3.95	332,+14			11.7	188,+21		
8	ch	6.8	179,+28			2.40	358,01			3.06	001,−06	9.0	179,+21	0.72	191,+24
9A	th	11.9	174,+21	2.25	002,+00			2.97	001,−05			14.0	175,+18		
9B	ch	12.0	176,+17			1.62	352,+13			2.28	354,+03	13.4	176,+14	0.75	179,+20

TABLE 2 (concluded)

Specimen	treatment	NRM Mi	NRM D°,I°	RV: DRM+CRM$_A$ THERMAL Mi	RV: DRM+CRM$_A$ THERMAL D°,I°	RV: DRM+CRM$_A$ CHEMICAL Mi	RV: DRM+CRM$_A$ CHEMICAL D°,I°	CRM$_A$ THERMAL Mi	CRM$_A$ THERMAL D°,I°	CRM$_A$ CHEMICAL Mi	CRM$_A$ CHEMICAL D°,I°	CRM$_B$ Mi	CRM$_B$ D°,I°	DRM Mi	DRM D°,I°
							SITE V (cont)								
10A	ch	14.0	181,+20			0.30	240,+01			0.61	347,−12	13.9	180,+20	0.75	190,+10
10B	th	10.5	191,+21	3.86	310,−05			4.29	319,−06			12.8	175,+19		
11	ch	10.9	182,+20			1.72	325,+02			2.24	336,−03	12.2	177,+17	0.66	186,+15
MEAN		*10.4*	*187,+25*	*3.12*	*342,+02*	*1.20*	*325,+13*	*3.82*	*345,−02*	*1.81*	*348,−06*	*12.6*	*180,+21*	*0.81*	*179,+15*
(*k*)			(32)		(11)		(3)		(17)		(53)		(145)		(44)
							SITE VI								
1	ch	10.1	176,+26			0.34	254,−16			0.89	338,−33	10.1	174,+27	0.99	180,+27
2	th	12.0	168,+25	2.10	341,−32			3.19	348,−28			14.1	167,+26		
3A	ch	12.0	170,+28			0.44	206,−04			0.93	352,−31	11.7	169,+29	1.27	184,+21
3B	ch	10.8	169,+25			0.78	278,−01			1.59	329,−17	11.1	165,+24	1.27	179,+21
4A	ch	9.4	170,+20			0.54	277,−14					9.6	167,+20	1.01	178,+14
4B	th	11.4	164,+27	2.60	347,−27			3.58	350,−23			14.0	165,+27		
5	th	5.7	153,+34	3.50	352,−37			4.60	354,−33			9.1	160,+35		
MEAN		*10.2*	*167,+27*	*2.73*	*347,−32*	*0.53*	*255,−10*	*3.79*	*351,−28*	*1.16*	*337,−25*	*11.4*	*167,+27*	*1.14*	*180,+21*
(*k*)			(116)		(142)		(6)		(205)		(42)		(187)		(191)
							SITE VII								
1A	th	20.5	178,+19	2.65	174,+18			1.59	171,+19			17.9	179,+19		
1B	ch	12.5	177,+25												179,+16
2A	ch	11.8	176,+24												166,+25
2B	th	11.5	174,+27	2.28	168,+15			1.24	170,+06			9.3	176,+30		
3A	ch	17.6	175,+16												170,+22
3B	th	13.1	180,+15	3.18	168,+10			2.15	167,+04			10.0	184,+17		
4A	th	11.8	176,+18	2.58	172,+13			1.53	168,+18			9.2	177,+19		
4B	ch	22.0	175,+13												178,+06
5A	th	12.8	173,+17	4.55	170,+10			3.48	170,+10			8.3	175,+21		
5B	ch	11.3	174,+22												171,+10
6A	th	13.3	170,+22	2.78	175,+14			1.73	172,+19			10.6	169,+24		
6B	ch	14.9	169,+17												179,+06
MEAN		*14.4*	*175,+20*	*3.00*	*171,+13*			*1.95*	*170,+13*			*10.9*	*177,+22*	*(1.07)*	*174,+14*
(*k*)			(238)		(366)				(131)				(152)		(69)

NOTES:
Mi is the intensity in 10^{-6} emu; D and I are the declination east and inclination positive downward; *k* is the precision estimate (Fisher 1953).
DRM is the magnetization remaining after more than 4500 hr of leaching.
RV: DRM + CRM$_A$—THERMAL is the magnetization remaining after thermal demagnetization at 674 °C.
CRM$_A$—THERMAL is obtained by vector subtraction of DRM from RV: DRM + CRM$_A$.
CRM$_A$—CHEMICAL is the most northerly directed vector obtained by vector subtraction of DRM from the chemical curve (see Fig. 2).
RV: DRM + CRM$_A$—CHEMICAL is the vector on the chemical curve corresponding to the CRM$_A$ vector.
CRM$_B$ is obtained by vector subtraction of RV: DRM + CRM$_A$ from NRM.
*These specimens were started in 10 N HCl and the transition point between the CRM$_A$ and CRM$_B$ could not be determined.

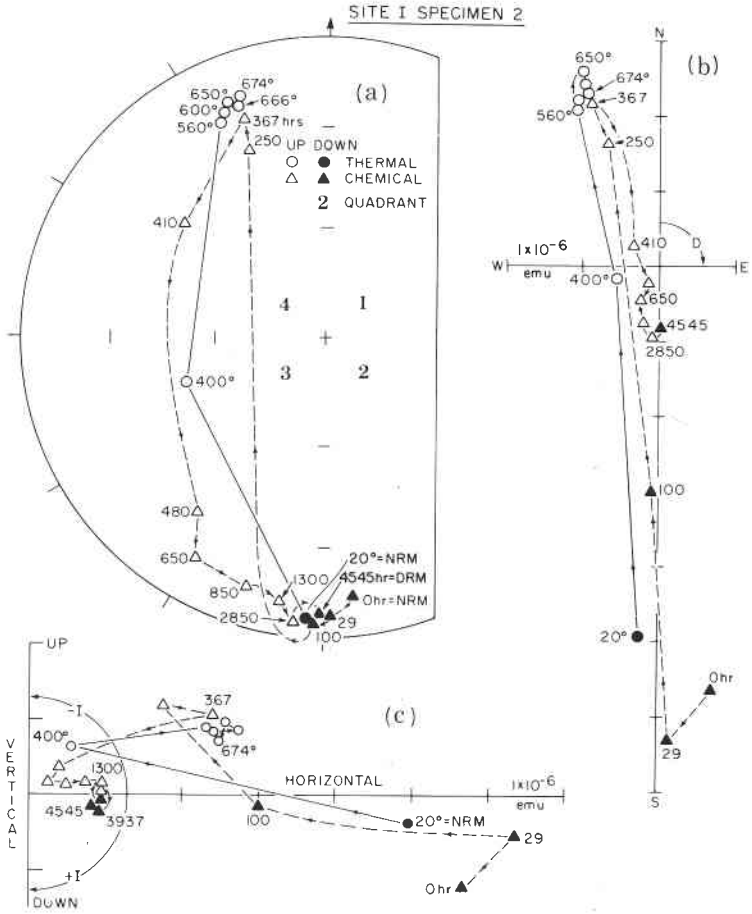


FIG. 1. Thermal demagnetization and chemical leaching of two specimens from the same core. Chemical leaching shows that the southerly directed NRM is composed of three magnetizations represented by the resultant vector RV: **DRM** + **CRM**$_A$ + **CRM**$_B$; the DRM and CRM$_B$ being southerly directed and the CRM$_A$ northerly directed. Upon leaching for 367 hr the CRM$_B$ is removed yielding the northerly directed resultant vector RV: **DRM** + **CRM**$_A$. Upon further leaching, the CRM$_A$ is removed and the southerly directed DRM is uncovered (4545 hrs). The **CRM**$_B$ is also removed by thermal demagnetization at 674 °C leaving the northerly directed RV: **DRM** + **CRM**$_A$ as indicated by the similarity of the 367 hr and the 674 °C results. The equal area polar projection stereogram (*a*) shows the direction changes occurring during thermal demagnetization (shown in °C) or chemical leaching (shown in hours—all numbers without the superscript sign); open (closed) symbols indicate negative (positive) inclinations. The circles and triangles in (*b*) show, in successive positions, the end point of the horizontal component of the magnetic vector obtained after the indicated treatment; the distance from the origin indicating the length of the vector component, that is, the intensity of the horizontal component; and the position with respect to the north axis indicating the declination. Part (*c*) shows the end point of the vector as a function of the horizontal and vertical components. The distance from the origin is a measure of the length of the vector or total intensity of magnetization. The inclination of the magnetic vector is readily obtained from the angle between the vector (whose endpoint is shown) and the horizontal. Between 700 and 4545 hr, all chemically leached specimens (except for site IV) were measured 11 to 14 times at intervals of 100 to 600 hr. Only a few of those results are shown on this (and following) figure to facilitate the reading of the graphical representations (the omitted results fall on or near the curves drawn). For the same reason, some of the results are not shown in all three (*a*), (*b*), and (*c*) figures but only in one or two of them. The unit in all vector diagrams is standardized to 1×10^{-6} emu although the size of the diagrams varies.

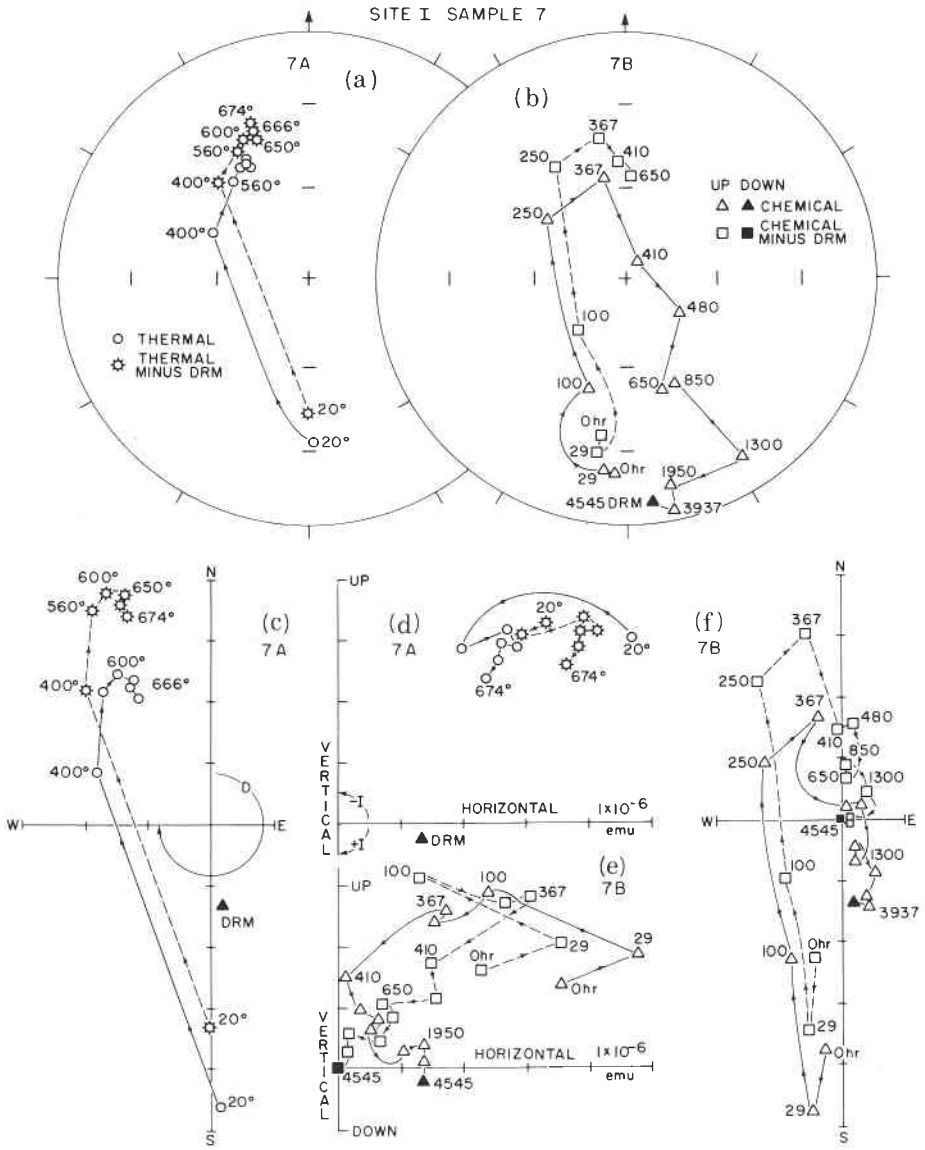


FIG. 2. Vector representation of thermal demagnetization (7A) and chemical leaching (7B) results obtained before and after removing the magnetization attributed to the DRM. The thermal minus DRM and the chemical minus DRM are obtained by subtracting the DRM (4545 hr) vector from the thermal and chemical curves respectively. The fact that the direction remains nearly constant (*a*) between 600 and 674 °C indicates that the CRM_B has been demagnetized completely so that the direction of the thermal minus DRM curve (8–wing circles) in that range can be attributed to the CRM_A solely. The reduction in intensity between 600 and 674 °C (*d*) indicates that a small fraction of the CRM_A may be demagnetized in that range. The chemical minus DRM curve (squares) represents the CRM_B–CRM_A curve with the 367 hr result as the estimated transition point between the southerly directed CRM_B and the northerly directed CRM_A. The agreement between the 367 hr and 674 °C vectors suggests that there is little overlapping of the solubility spectra of the CRM_A and CRM_B; that is, at the transition point, most of the CRM_B has been leached out while most of the CRM_A remains intact. The directions (squares in *b*) past 650 hr are not plotted, since, as the curve (squares in *e* and *f*) tends towards zero, the directions become less and less significant owing primarily to the possible acquisition of small viscous components (see footnote 1).

towards the north until both the direction and the intensity are in close agreement with those obtained after demagnetization at 674 °C. Upon further leaching, the vector then migrates southwards until it remains constant[1] for the last 2000 to 3000 hr of leaching. It is suggested that chemical leaching successively removes two magnetizations of chemical origin: First a southerly directed CRM_B, and later, a northerly directed CRM_A. The fact that both the direction and intensity remain unaffected by leaching (no vector change for the last 2000 to 3000 hr) indicates that all magnetization of chemical origin has been removed and that the remaining magnetization is a simple DRM.

The similarity between the vectors obtained after partial leaching (367 hr) and after demagnetization at 674 °C (Fig. 1) indicates that, in both instances, the northerly directed magnetization is composed of a southerly directed DRM and a northerly directed CRM_A of greater intensity. The northerly directed CRM_A vector can be obtained by subtracting the DRM vector from the northerly directed resultant vector (RV: **DRM** + $\mathbf{CRM_A}$) as shown in Fig. 2.

The results obtained by performing this vector subtraction on all specimens (except specimen 4) of site I are given in Table 2 under the heading CRM_A. For a thermally demagnetized specimen, the DRM vector (Table 2, last column) obtained from the chemically-treated paired (A or B) specimen is subtracted from the 674 °C RV: **DRM** + $\mathbf{CRM_A}$; where there is no paired specimen (specimen 3), the site mean DRM vector is then used for the subtraction.

For a chemically leached specimen, the CRM_A vector is determined according to the following procedure illustrated in Fig. 2*b*, *e*, and *f*. First the southerly directed DRM is removed by subtracting the DRM vector (the 4545 hr result) from the points on the chemical leaching curve. The resultant curve (chemical minus DRM; square symbols) represents the trace of the vector composed of CRM_A and CRM_B. The most northerly direction indicates the point of transition between the southerly directed CRM_B and the northerly directed CRM_A. The vector representing this point of transition is listed in Table 2 under the heading CRM_A—CHEMICAL.

Laboratory experiments in the earth's field on untreated rocks indicate that, at room temperature, the NRM of these red beds is stable; that is, the specimens do not acquire any non-negligible viscous magnetization. The NRM is then represented by the resultant vector (RV: **DRM** + $\mathbf{CRM_A}$ + $\mathbf{CRM_B}$), and the CRM_B vector (Table 2) is then obtained by subtracting RV: **DRM** + $\mathbf{CRM_A}$ from the NRM vector.

Accuracy of vector determination

In all the thermal results from this site, the directions change very little (Figs. 1*a*, 2*a*) between 600 °C and 674 °C indicating that the CRM_B has been completely demagnetized. In some specimens (Fig. 2*d*), the intensity decreases between 600 °C and 674 °C indicating that RV: **DRM** + $\mathbf{CRM_A}$ reduces in that range. If the DRM vector is taken as a constant, this means that, when using the 674 °C result, the vector length is, in some instances, too short and the calculated intensity is less than the true CRM_A intensity. However, because of the near anti-parallelism (154°) of the DRM and CRM_A vectors (site means given in Table 2), the use of a slightly-short (in length) CRM_A vector does not affect its direction appreciably. For example, in specimen 7A (Fig. 2), although the length of the CRM_A vector reduces by 13% from 600° to 674 °C (Fig. 2*d*), its direction (Fig. 2*a*) for each of the 600, 650, 666, and 674 °C points remains within ±2° of arc. Therefore, the CRM_A (THERMAL) intensities (Table 2) should be regarded as

[1]The small changes observed between 1300 and 4745 hr are mostly attributable to small viscous components. Experiments on the acquisition and removal of viscous components are discussed later. However, it should be mentioned that, because of problems arising from acid fumes, the chemical experiments are performed in a specially built fiberglass hut where no field cancellation is available. Small viscous components (of intensity of 10^{-8} to 10^{-7} emu) acquired in the earth's field are sometimes (depending on the ratio of viscous/stable magnetizations) reflected in the observations obtained after 800 hr of leaching when the intensity of the stable magnetization is small. However, for the last result (>4500 hr) shown in the figures and the tables, the viscous magnetization has been removed by leaving the specimens in a zero-field for a sufficiently long period of time before measurement.

minimum values which might be too low by several percent, though the directions are probably accurate to within a few degrees. Likewise, for the determination of the CRM_B vector, inaccuracies resulting from the use of the 674 °C results will be reflected more in the intensities than in the directions, since the CRM_A and the CRM_B are also nearly anti-parallel (157°). Furthermore, because the intensity of the CRM_B is two to three times greater than that of the CRM_A (Table 2), an error introduced by using a 'too-short' CRM_A vector is that much reduced. Thus the accuracy (of both intensity and direction) of the CRM_B is relatively greater than that of the CRM_A.[2]

In the chemical experiments, it is generally more difficult to determine accurately the CRM_A vector. This is because (*1*) the point of transition between the CRM_B and the CRM_A may occur during a leaching period,—that is, between two measurements,—and (*2*) the solubility (in HCl) spectrum of each CRM, although they are on the whole clearly different, may overlap slightly. Because the CRM_B and the CRM_A are nearly anti-parallel, inaccuracies owing to (*1*) and/or (*2*) will be reflected (as with the thermal results) much more in the magnitude than in the direction of the CRM_A vector.

However, at this site, both the intensity and the direction of the CRM_A (THERMAL) and the CRM_A (CHEMICAL) are closely similar (Table 2) indicating (*1*) that the 367 hr time mark is very near the transition point, and (*2*) that there is little overlapping (if any) of the solubility spectra of the CRM_A and CRM_B. It then appears that the CRM_B ceases to decrease before the CRM_A starts to decay and that the CRM_A (CHEMICAL) vector is as accurately determined as the CRM_A (THERMAL) vector.

[2]The CRM_A and CRM_B intensities would probably be more accurate if, for each specimen, the largest intensity obtained at or after 600 °C was used for vector subtraction. However, since slight inaccuracies in intensities do not appreciably affect the directions of magnetization, more accurate determinations are not required for this study. Consequently, because standardization simplifies the presentation, 674 °C, which is the highest cleaning temperature for 80% of the thermally treated specimens is adopted as the standard.

Field or self reversals?

The high precision of the DRM ($k = 208$) and the CRM_B ($k = 91$) leaves little doubt that both were acquired in a southerly directed field. The oppositely directed CRM_A could be attributed to a reversal of the earth's field or to self-reversals of some of the magnetic grains. All the results of this study can be explained in terms of field reversals, but many features (which shall be discussed as they are observed) cannot be explained if self-reversal is invoked. For example, at this site, the streaking of directions towards the south (Fig. 3) may be the recording of the first stage of a field reversal. This is evidenced by the relative positions of the specimens, with the stratigraphically lower specimens 1 and 2 being the nearest to anti-parallelism with respect to the DRM and the CRM_B. With the self-reversal hypothesis, it is difficult to explain why the direction of magnetization of specimen 7 diverges more from anti-parallelism (035°) than does specimen 1 (015°). The 20° displacement of direction which occurs over the 1.55 m stratigraphic thickness sampled indicates that a complete field reversal might be recorded in the 14 m of sediments adjacent to this site. However, since the rate of acquisition of the CRM_A is not known, the time taken for the field to reverse cannot be determined from this result.

It is worth noting that there is *no evidence* of a lowering of intensities as the inclination becomes steeper (Table 2). This surprising result might be interpreted as an indication that the field intensity does not reduce during a reversal. More intermediately-directed (near the east–west demarcation line) magnetizations are required to ascertain this. Intermediate directions during field reversals were found by Vlasov and Kovalenko (1963). Although the authors do not analyse their results in terms of intensity, inspection of Fig. 5 of their paper shows no evidence that the intermediately-directed magnetizations are systematically weaker than the magnetizations acquired before and after the field reversals. It is usually difficult to relate intensity of magnetization with field intensity, because the proportion of the magnetic carriers may vary greatly from specimen to specimen. However, at site I, the intensity of each magnetization—DRM,

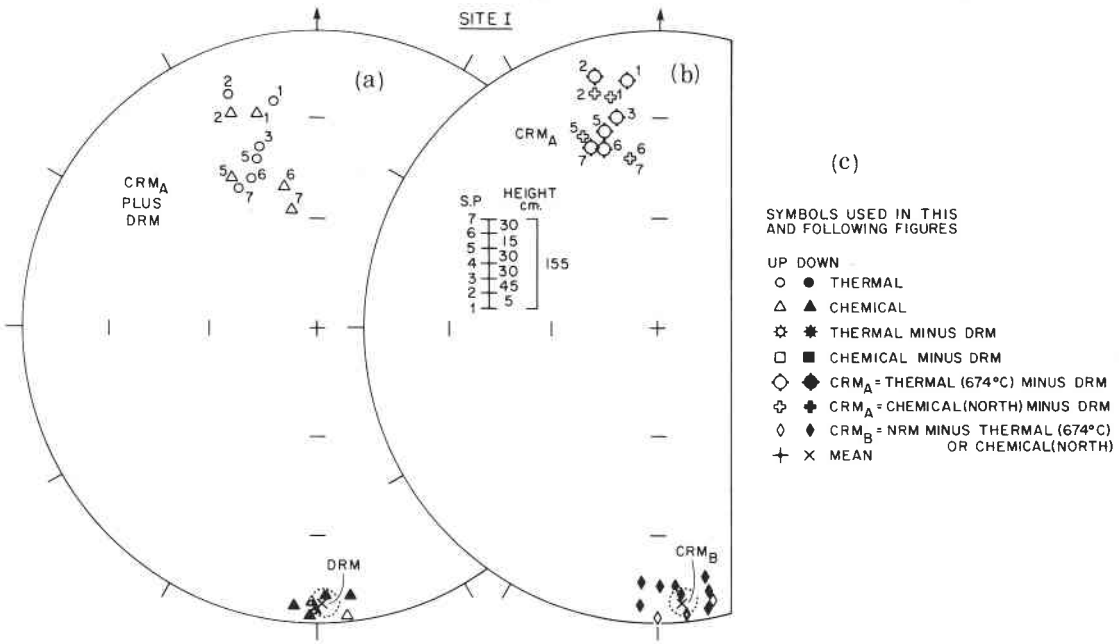


FIG. 3. Directions of magnetization of the three magnetizations contained in the specimens of site I. The DRM and CRM_B site-means are shown with their α_{95} circles of confidence. Each CRM_A in (*b*) is obtained by subtracting the DRM vector of that core from the RV: **DRM** + **CRM_A** given in (*a*). It is suggested according to the relationship between the streaking of directions towards the south and the relative stratigraphic heights of the specimens that the CRM_A was acquired at the beginning of a field reversal. Interpretation of symbols (*c*): circles and triangles are directions and vector end points obtained directly (no vector analysis) from the thermal and chemical experiments respectively as shown in Fig. 1. The last triangle obtained after more than 4500 hr of leaching and subsequent removal of viscous components is called DRM. That DRM subtracted from the thermal and chemical results yields the thermal minus DRM (8–wing circles) and the chemical minus DRM (squares) curves as shown in Fig. 3. The result obtained at 674 °C on that thermal minus curve is taken to represent the CRM_A vector and the 4–wing large circle is then used (Fig. 4*b*). The most northerly direction (transition point) of the chemical minus DRM curve is taken to represent the CRM_A vector and the coarse cross is then used (Fig. 4*b*). The diamonds represent the CRM_B vectors obtained by vectorially subtracting, from the NRM, RV: **DRM** +**CRM_A**—that is, the 674 °C thermal result or the result on the chemical curve yielding the transition point.

CRM_B, and CRM_A (thermal and chemical)—does not vary much from specimen to specimen (Table 2) indicating that the composition of all specimens is fairly uniform and that the intensities may be related to field intensity. It is then suggested that these results signify that the field began to reverse by changing its direction without reducing its intensity.

Sites II and III

The three magnetizations were acquired in a southerly directed field (Fig. 4*a*, 4*b*). Comparison of the mean directions (Table 2) using the *F*-test (Watson 1956) shows that there is no significant difference between the mean directions of all the different magnetizations at these two sites except for the CRM_A of site II. At that site, comparisons of CRM_A with DRM and CRM_A with CRM_B yield respective values of 23.4 and 18.9 which are much greater than the respective values $F_{.005}$ (2,26) = 6.5 and $F_{.005}$ (2,28) = 6.4 given in the *F*-tables. Therefore, even at the .005 level the null hypothesis does not hold, thus indicating that the direction of the CRM_A (site II) is significantly different from those of the DRM and CRM_B. This difference may be attributable to secular variation, or to the CRM_A

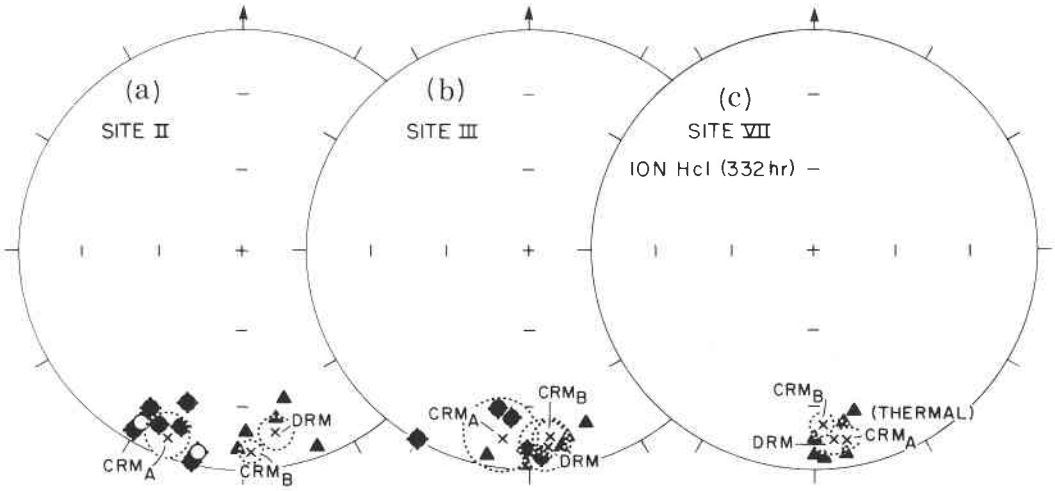


FIG. 4. Mean site directions and α_{95} circles of confidence of the three magnetizations. The CRM_Bs at the three sites, the CRM_As at site VII and two DRMs (170°, +17°) at site II are not shown because of insufficient space; however, those directions are given in Table 2.

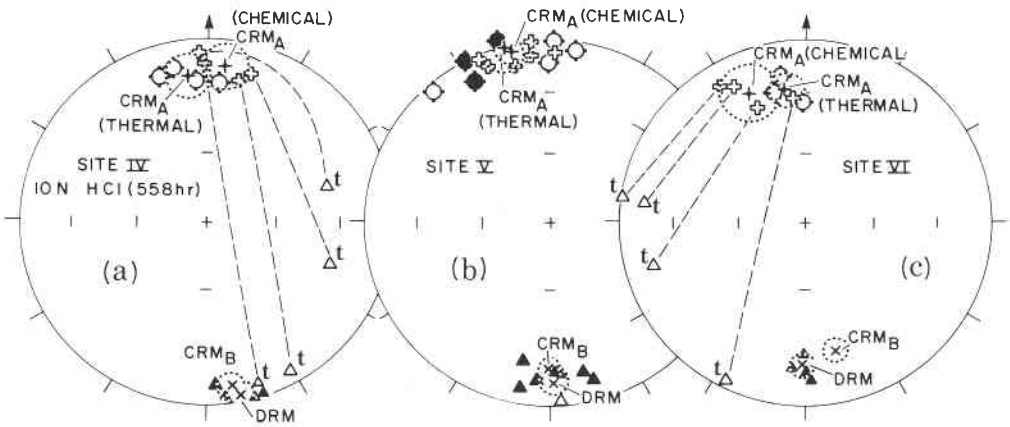


FIG. 5 The CRM_A at these three sites is northerly directed as shown by the thermal and chemical results with intersecting circles of confidence. In (*a*) and (*c*), two directions (per specimen) obtained during chemical leaching are shown: one (triangle) obtained after 4545 hr belongs to the tight DRM grouping; the other (triangles with superscript t) is the direction which, when the DRM vector is subtracted from the chemical curve yields the most northerly direction (transition point) represented by the coarse cross. Those two directions (before and after vector subtraction) are joined by a dashed line. The leaching time at which the transition point is obtained varies from site to site and amongst specimens. It is 232 hr for site IV, 340–540 hr for site V and 250–340 hr for site VI. All specimens of site IV were leached in 10N HCl from the beginning (0 hr) to the end (558 hr) of the experiment.

being acquired near the beginning or the end of a field reversal. The latter is more probable owing to the fact that there is no streaking of directions towards the CRM_B or DRM site-mean direction. Because most of the CRM_A and CRM_B magnetizations are southerly directed, the transition point cannot be established and therefore the CRM_A (CHEMICAL) vector could not be obtained at these two sites (Table 2).

At three (II, III, and VII) of the first seven sites, the CRM_A is southerly directed; at the other four sites, it is northerly directed. The specimens of all these sites appear identical in

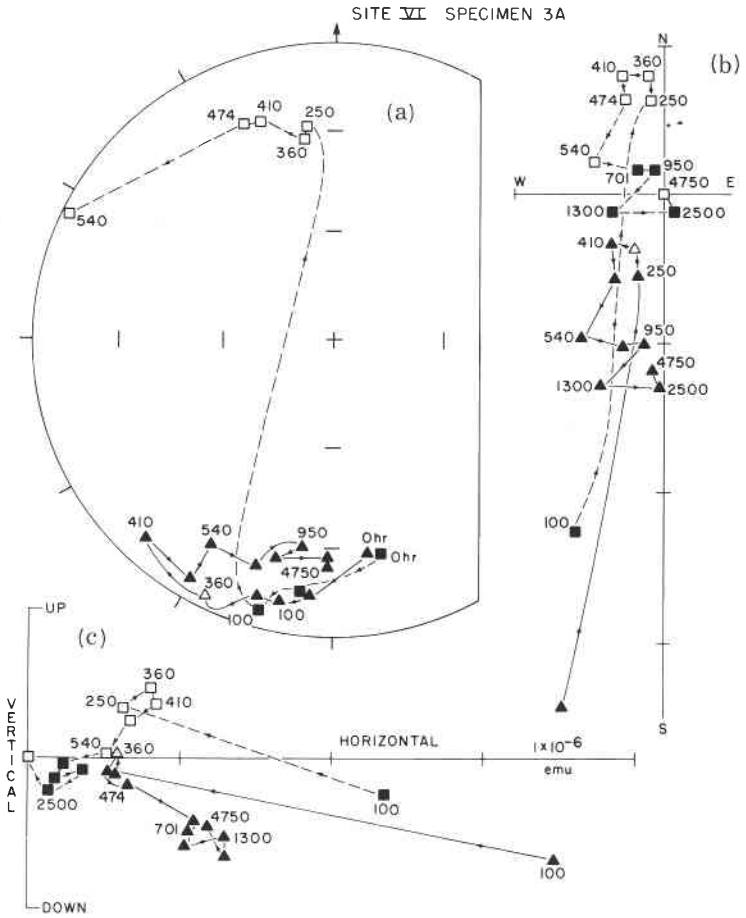


FIG. 6. Vector diagram of chemical leaching of a specimen of site VI showing the result of vector analysis. The northerly directed magnetization is not evident in the chemical curve (triangles); (*a*) showing the directions remaining well south in the third quadrant. However, (*b*) and (*c*) show that the 250–474 hr vector triangles are much smaller than the 4750 (DRM) vector indicating the presence of two magnetizations. When the DRM vector is subtracted (open squares), it is shown that the CRM_A is definitely northerly directed. This example shows the importance of drawing the chemical minus DRM (squares) vector diagrams to determine the direction of the CRM_A and the transition point (360 hr in this instance).

composition. This again favours a field reversal rather than a self reversal as the cause of the northerly directed CRM_A.

Sites IV, V, and VI

The CRM_A at these three sites is northerly directed (Fig. 5). The intensity of the CRM_A (CHEMICAL) is substantially lower (unlike that of site I) than the intensity of the CRM_A (THERMAL) (Table 2). Because 10 N HCl was used for leaching the specimens of Site IV, the leaching rate was rapid and it is quite possible that the transition point occurred between two measurements at that site. However, at Sites V and VI, the directions usually remain northerly directed for several measurements (Fig. 6, 250–474 hr). This indicates (*1*) that the transition point was not missed, and (*2*) that the solubility spectra overlap. These low intensities are then attributed to overlapping mostly. The fact that these intensities are low and comparable to the DRM intensities (Table 2) accounts for the high scatter of directions (Fig. 5, triangles with superscript t = RV: **DRM** + **CRM_A**) obtained directly from the chemical results. However, the close grouping

obtained after subtracting the DRM vector shows that a good estimate of the CRM_A direction can be calculated by vector analysis. The improvement in grouping is well shown by the ratio k (CRM_A) over k (RV: **DRM** + **CRM_A**) calculated from the Table 2 results; the ratios being 17, 17, and 7 for sites IV, V, and VI respectively. Although the RV: **DRM** + **CRM_A** directions obtained thermally (Table 2) are much less scattered than those obtained chemically, their grouping is likewise improved after vector analysis; the k ratio being 2 for site IV and 1.5 for sites V and VI.

It is concluded that either of the two methods,—a combination of thermal (674 °C) and chemical (DRM) experiments or chemical experiments solely,—could be used for determining (at these three sites) the direction (but not the intensity) of the CRM_A with a reasonable degree of accuracy, even though the solubility spectra of the CRM_B and CRM_A apparently overlap. The averages of the thermal and chemical results for the three site-mean directions are 349°, −17° and 351°, −16° respectively. No significant difference could be found between the CRM_B of the thermally treated specimens and that of the chemically treated specimens, so they have been compiled together.

Site VII

Fresh 10N HCl was used to leach these specimens. After 100 hr, the specimens began to expand radially, then separated (233 hr) into circular layers about the axis and finally (322 hr) the outer layers crumbled. This unusual effect was not observed when another specimen of this site and a specimen of site IV were immersed in one year old 10N HCl. Hence the acid, as expected, loses its concentration with time, thus making it more difficult to control chemical experiments. This is the reason for leaching the other sites in 8N HCl for the first 1300 hr. Because of the irregular shape of the final remains of these specimens, the intensity could not be determined accurately. In the vector analysis for the determination of the CRM_A and CRM_B vectors, the mean of the DRM intensities (1.07×10^{-6} emu) of the first six sites has been used. Since all three magnetizations are southerly directed, the determination of the CRM_A and CRM_B directions is little affected by the possible inaccuracy of the DRM intensity thus calculated.

Transition point

Several additional specimens from sites I, IV, and V were chemically leached, and sliced at different stages of leaching to observe the effect of acid penetration with time. There is a definite relationship between the northward progress of the direction and the gradual disappearance of the red coloring, presumably owing to deeper penetration of the acid, and the point of transition appears to coincide with complete removal of the red coloring throughout the specimen. This evidence, combined with the study of polished sections described later, indicates that the CRM_B is carried by the red cement.

Sites VIII, IX, X, and XI

The site mean NRM directions at these four sites, which are all within 4.5 m of stratigraphic height (Appendix), are southerly directed (Table 3). At three sites (IX, X, and XI) the precision is very poor. Under thermal or chemical treatment, some directions remained south, and some began to move northwards. However, with few exceptions, the northward movement stopped at an intermediate direction (Fig. 7).

In most of the specimens, the index of within-specimen uniformity of direction ($\Delta\Psi$, Roy and Fahrig 1973) was large (Table 3). This index is obtained using the empirical formula

$$[1] \quad \Delta\Psi = [((\Delta D \cos I_1)^2 + \Delta I_1^2)/2]^{1/2}$$

where ΔD is the angular difference between the declination obtained with the specimen first in an upright and then in an inverted position under the astatic magnetometer; ΔI_1 is the angular difference between the inclinations measured with the specimen in two positions about its axis and positioned so that the mean horizontal component rotates in a horizontal plane; and I_1 is the mean inclination with respect to the specimen. A large $\Delta\Psi$ indicates that the direction of magnetization within the specimen is not uniform. So, some[3] of the

[3]The importance of $\Delta\Psi$ was not known at the time of sampling, so that the specimens were not necessarily cored perpendicular to the bedding but from the most easily accessible direction. If $\Delta\Psi$ is an expression of time of magnetization within a specimen, then it is preferable to core perpendicular to bedding. In this collection, the specimens whose coring axes were the nearest to the perpendicular of the bedding plane were chosen for slicing.

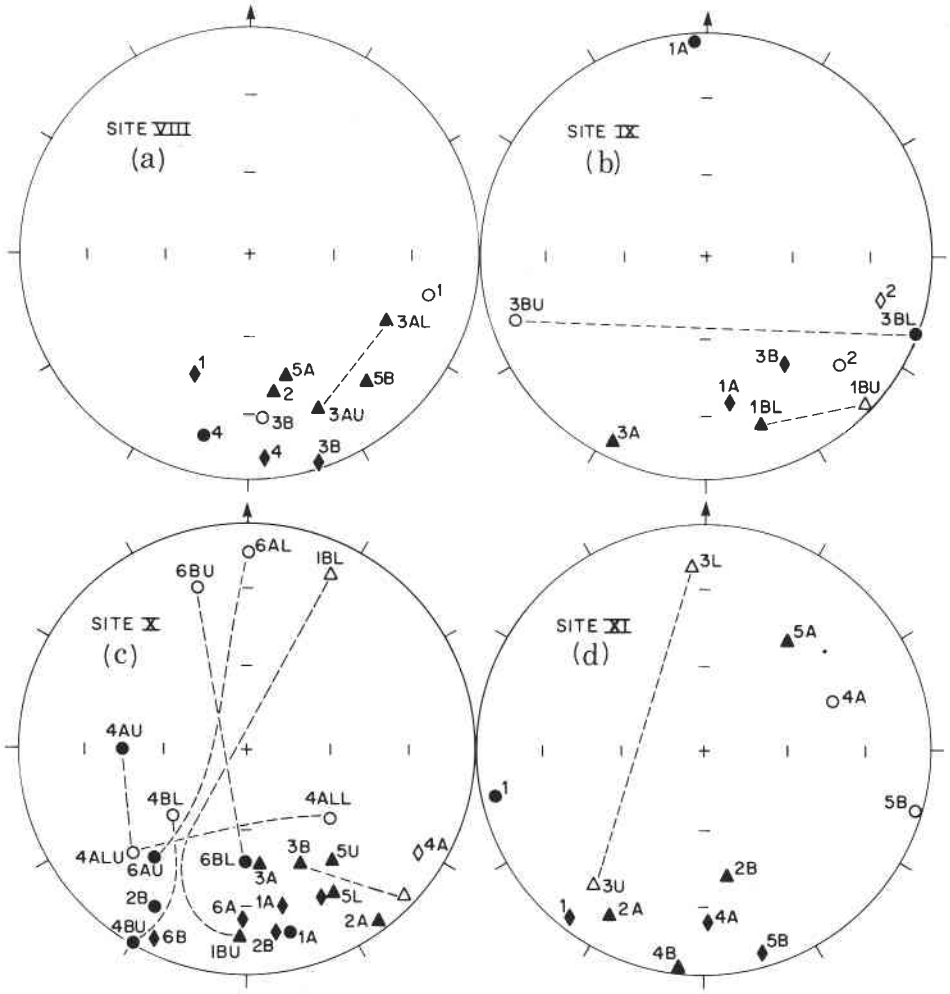


FIG. 7. Directions of magnetization of DRM (triangles), CRM_B (diamonds), and RV: **DRM** $+$**CRM**$_A$ (circles) obtained at four sites within 4.5 m of stratigraphic height (Appendix). The dashed lines join the two halves (U for upper half and L for lower half) of specimens which were cut at some stage during treatment (more details are shown in Figs. 8 and 9). Specimen 4 of site X was cut into an upper half and two lower (upper (LU) and lower (LL)) quarters. Intensities corresponding to each direction are given in Table 3.

specimens (with a large $\Delta\Psi$) whose directions stabilized during thermal or chemical treatment at an intermediate direction, were cut into halves and the treatment resumed. As expected, the direction of each half is different from that of the whole specimen (Figs. 8 and 9) and the $\Delta\Psi$ is smaller (Table 3).

In general, one of the halves has a direction near the northerly or southerly direction found in sites I–VII (Fig. 8*b*, 8*c*); sometimes, the directions of both halves (Figs. 8*a*, 9) are near those directions. In some instances, the north (or south) direction is reached immediately after cutting (Fig. 8*a*, lower half); in other instances, it is reached only after further leaching (Fig. 8*a*, upper half). Although different specimens behave differently, it is believed that the two halves were magnetized in fields of opposite polarities and that the DRM at these sites was acquired during a field reversal. The fact that some directions do not quite reach the south (or the north) direction (Fig. 8*c*, upper half) is not contrary to this view as those particular halves probably contain two magnetizations of opposite polarities as the original whole specimens evidently

TABLE 3. Chemical leaching and thermal demagnetization results of the specimens of sites VIII–XI

Specimen	treatment	NRM Mi	NRM D°,I°	RV: DRM + CRM_A – THERMAL Mi	D°,I°	$\Delta\Psi_1$	$\Delta\Psi_2$	CRM_B Mi	CRM_B D°,I° (*k*)	DRM Mi	DRM D°,I°	DRM $\Delta\Psi_1$	DRM $\Delta\Psi_2$
						Site VIII							
1	th	7.5	179,+36	2.78	102,−21	56		8.13	204,+42				
2	ch	5.7	160,+07							0.65	170,+38	23	
3A	ch	6.7	167,+25							1.50	156,+26U	31	4
										2.20	116,+33L		26
3B	th	8.1	166,−05	1.83	185,−28	09		6.56	160,+01				
4	th	8.8	180,+13	2.53	193,+19	38		6.40	175,+10				
5A	ch	7.4	161,+22							1.20	137,+25	04	
5B	ch	5.4	172,+20							0.81	163,+43	20	
MEAN								*7.03*	*178,+18 (8)*				
						Site IX							
1A	th	2.5	167,+44	0.69	356, 05	11		2.95	170,+34				
1B	ch	5.7	159,−21							1.81	133,−04U	61	31
										1.55	162,+22L		16
2	th	10.7	111,−24	2.58	129,—25	40		8.28	106,−23				
3A	ch	5.2	200,+28							2.15	206,+08	23	
3B	th	3.5	159,+26	4.49	251,−12U	157	59	2.96	145,+40				
				4.00	110,+01L		81						
MEAN								*4.73*	*139,+20 (3)*				
						Site X							
1A	th	6.7	167,+27	1.26	168,+17	37		5.44	167,+29				
1B	ch	4.7	166,−02							1.93	182,+15U	98	20
										0.25	026,−16L		36
2A	ch	5.3	182,−13							1.18	143,+01	23	
2B	th	7.7	177,+18	0.94	210,+20	33		6.94	173,+17				
3A	ch	7.1	178,+15							0.84	175,+47	55	
3B	ch	5.5	169,+29							3.04	134,−03U	63	34
										2.36	156,+42L		38

TABLE 3 (concluded)

Specimen	treatment	NRM Mi	NRM D°,I°	RV: DRM + CRM_A − THERMAL Mi	D°,I°	$\Delta\Psi_1$	$\Delta\Psi_2$	CRM_B Mi	CRM_B D°,I° (*k*)	DRM Mi	D°,I°	$\Delta\Psi_1$	$\Delta\Psi_2$
4A	th	7.2	155,−10	8.20	269,+45U	46	03	7.80	124,−09				
				14.30	227,−32LU		09						
				7.30	131,−50LL		21						
4B	th	4.8	177,+00	2.48	210,+00U	149	03	3.88	153,+25				
				4.80	227,−53L		84						
5	ch	6.7	164,+01							0.95	144,+37U	42	04
										4.60	151,+26L		25
6A	th	3.7	190,+34	1.89	223,+37U	119	62	4.92	182,+25				
				4.94	001,−14L		39						
6B	th	3.0	220,+15	4.80	343,−27U	77	25	2.97	207,+06				
				4.62	180,+48L		30						
MEAN								*5.33*	*168,+17* (7)				
						Site XI							
1	th	9.3	231,+06	3.43	256,+08	39		6.37	218,+04				
2A	ch	6.2	197,+42							0.56	208,+16	41	
2B	ch	8.1	133,+14							0.61	169,+40	37	
3	ch	6.1	192,−12							1.83	218,−23U	91	48
										1.80	355,−21L		12
4A	th	6.2	153,+05	3.48	070,−38	85		6.99	178,+23				
4B	ch	8.1	165,+16							0.77	185,+01	3	
5A	ch	5.2	162,+43							1.50	039,+40	46	
5B	th	8.9	154,+04	1.73	106,−02	34		7.88	163,+05				
MEAN								*7.08*	*188,+12* (7)				

NOTES: $\Delta\Psi_1$ and $\Delta\Psi_2$ are the index of within-specimen uniformity of direction before and after (respectively) cutting the specimens in halves. Other symbols as per Table 2. U and L indicate upper and lower halves of cut specimens. When the thermally treated specimens are cut, the CRM_B is obtained by subtracting, from the NRM vector, the *mean* of the RV: DRM + CRM_A of the half-specimens whose volumes are about only 0.41 the volume of the whole specimen because of the material lost during cutting. For specimen 4A, the lower half was cut in half: LU for the upper part and LL for the lower part. The volumes relative to that of the whole specimen are 0.41, 0.18, and 0.18 respectively. Code as per Table 2.

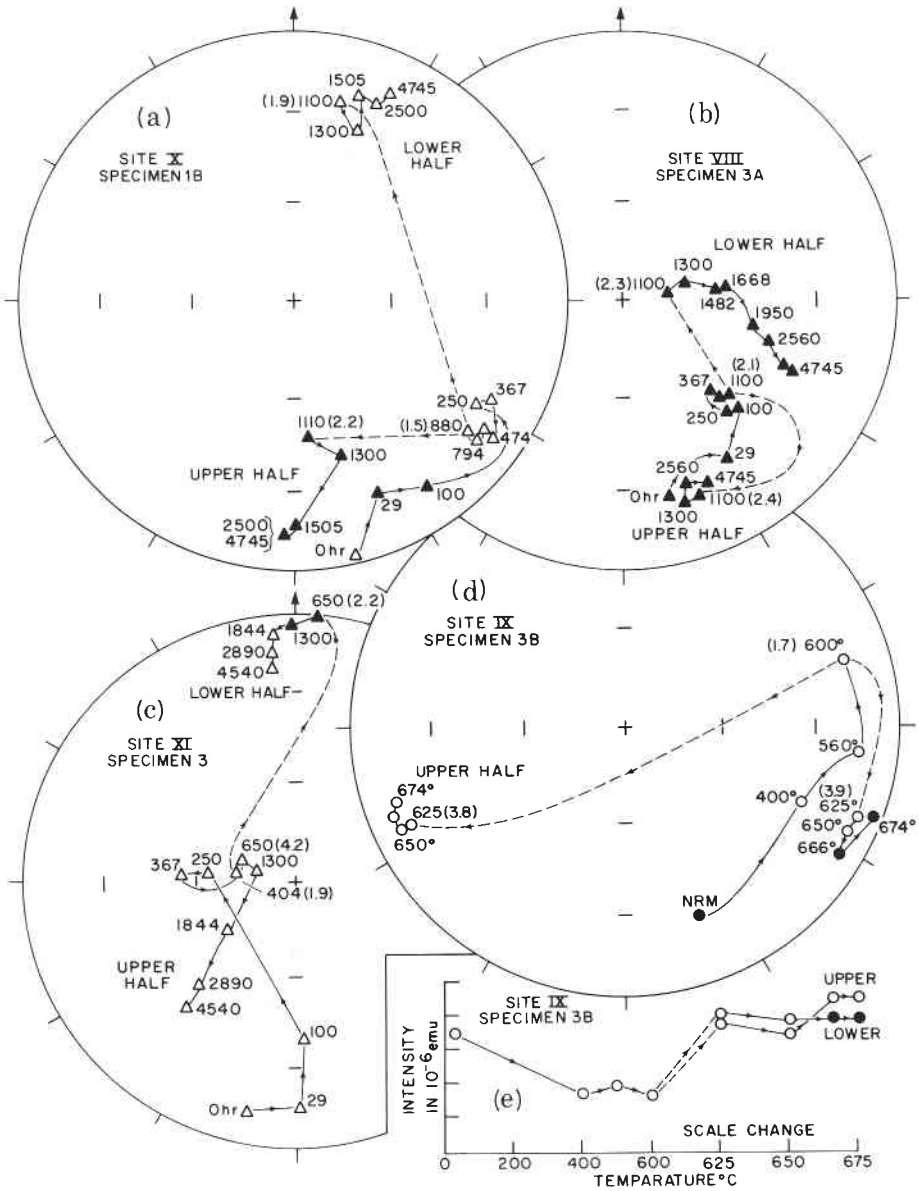


FIG. 8. Directions obtained during chemical (*a*, *b*, and *c*) or thermal (*d*) treatment. The specimens were cut into two halves at the time indicated (*a*, *b*, and *c*) and after the 600 °C heating, (*d*). The numbers in brackets indicate the intensities of the whole and half specimens obtained before and after cutting. It is noted that the intensity of each half is considerably larger than that of the whole specimen. The $\Delta\Psi$ after cutting is also smaller than before cutting (Table 3), although, in some instances (site IX, sp. 3B), it is still large. The relative intensities and $\Delta\Psi$ before and after cutting thus indicate that the upper and lower halves have been magnetized in opposite directions. It is suggested (see text) that these specimens acquired their magnetization along the north–south axis partly before and partly after a field reversal.

did. Unless the reversal occurred sharply and exactly where the specimen was cut, only one of the halves would be expected to reach the north (or south) direction.

Likewise, most of the directions obtained from the thermally demagnetized specimens tended towards the north or south direction (Fig. 9). Since, in general, the intensity of

RV: **DRM + CRM_A** is appreciably larger than that of the DRM (Table 3), it then appears that the CRM_A was also acquired during a field reversal. In rare instances, the directions of the two halves (Fig. 8*d*) did not move north and south but east and west. These directions could (*1*) record the directions of the earth's field during reversal or (*2*) be attributable to the resultant vector of two almost oppositely-directed magnetizations. Alternative (*2*) is more probable than (*1*) for the following reasons: (*a*) thermally treated specimens are composed of both DRM and CRM_A, (*b*) the intensities of the halves are almost invariably larger than those of the whole specimens (Figs. 8*e* and 9*d*), and (c) the $\Delta\Psi$ of an intermediately-directed half is usually large (Table 3). The (*b*) and (*c*) observations are also applicable to the chemically treated specimens. It is then suggested that both the DRM and the CRM_A were acquired along the direction of the field at that time (174°, +15°; 354°, −15°), and that most, if not all, intermediate directions are attributable to opposite magnetizations acquired along that direction.

The result obtained from sample 6 of site X (Fig. 9) indicates that field reversals may sometimes be of short duration. The upper part (6A upper) is southerly directed, the middle part (6A lower and 6B upper) is northerly directed, and the lower part (6B lower) is again southerly directed. It is possible, of course, that this sequence is owing to the resultant vector of a southerly directed DRM and a northerly directed CRM_A with a concentration of CRM_A-carrying grains in the middle. The intensity, however, does not support this assumption. The intensity of the (southerly directed) lower end (6B lower) is about the same as the intensity of the middle (6B upper and 6A lower) as shown in Fig. 9*d*. It appears then that this specimen has recorded polarity reversals of the CRM_A that could have been caused by two field reversals or by self-reversals. The latter is less probable, because it is difficult to explain why the middle but not the ends would be subjected to a self-reversing process. Although the rate of acquisition of the CRM_A is unknown, the fact that the north direction is confined to 2 cm of sediments suggests that the north polarity of the field was short-lived.

The CRM_B results (Fig. 7 and Table 3) indicate that, although, in some specimens, a small northerly directed magnetization may be present, most of the CRM_B (diamonds) was acquired when the field was southerly directed (only one direction (site IX, specimen 2) is found far away from the direction of the southerly directed field).

The results obtained at these four sites indicate the following:

(*1*) the field at that time was predominantly southerly directed; the number of south directions far outweighing the number of north directions,

(*2*) the field clearly reversed through 180° (Fig. 8*c*, lower half),

(*3*) the reversal was quite rapid; it can be recorded within a specimen (2.2 cm height) (Fig. 8*a*) and

(*4*) the reversal was of short duration (Fig. 9).

However, no relationship can be found between the stratigraphic levels and the polarity of magnetization (of either DRM or CRM_A) of the specimens (or parts of specimens). It is therefore uncertain whether or not the spot readings of the north directions obtained at different stratigraphic levels represent different reversals. Because of the presence of two magnetizations, different grain sizes of the DRM carriers and different periods (and timing rates) for the chemical process, it is possible that specimens at different levels acquired their northerly magnetization at about the same time, — that is, during the same reversal. A more interesting alternative is that these spot readings really represent different rapid and short-lived reversals. If so, the number of reversals (Fig. 7) that can have occurred in such a short time interval (4.5 m of deposition) indicates that, at reversal time, the field may have been subjected to an oscillatory motion before coming to rest in the same or opposite polarity.

Sites XII to XVII

Except for site XV and specimen 5B of site XIV, the CRM_A of all specimens is northerly directed (Fig. 10). An estimate of the CRM_A–CHEMICAL could not be obtained. The specimens are medium-grained and it appears that both CRM_A and CRM_B are rapidly leached out. However, there is evidence that the CRM_A–CHEMICAL is also northerly directed. In the early stage of leaching, a northward excursion of the direction is observed in most

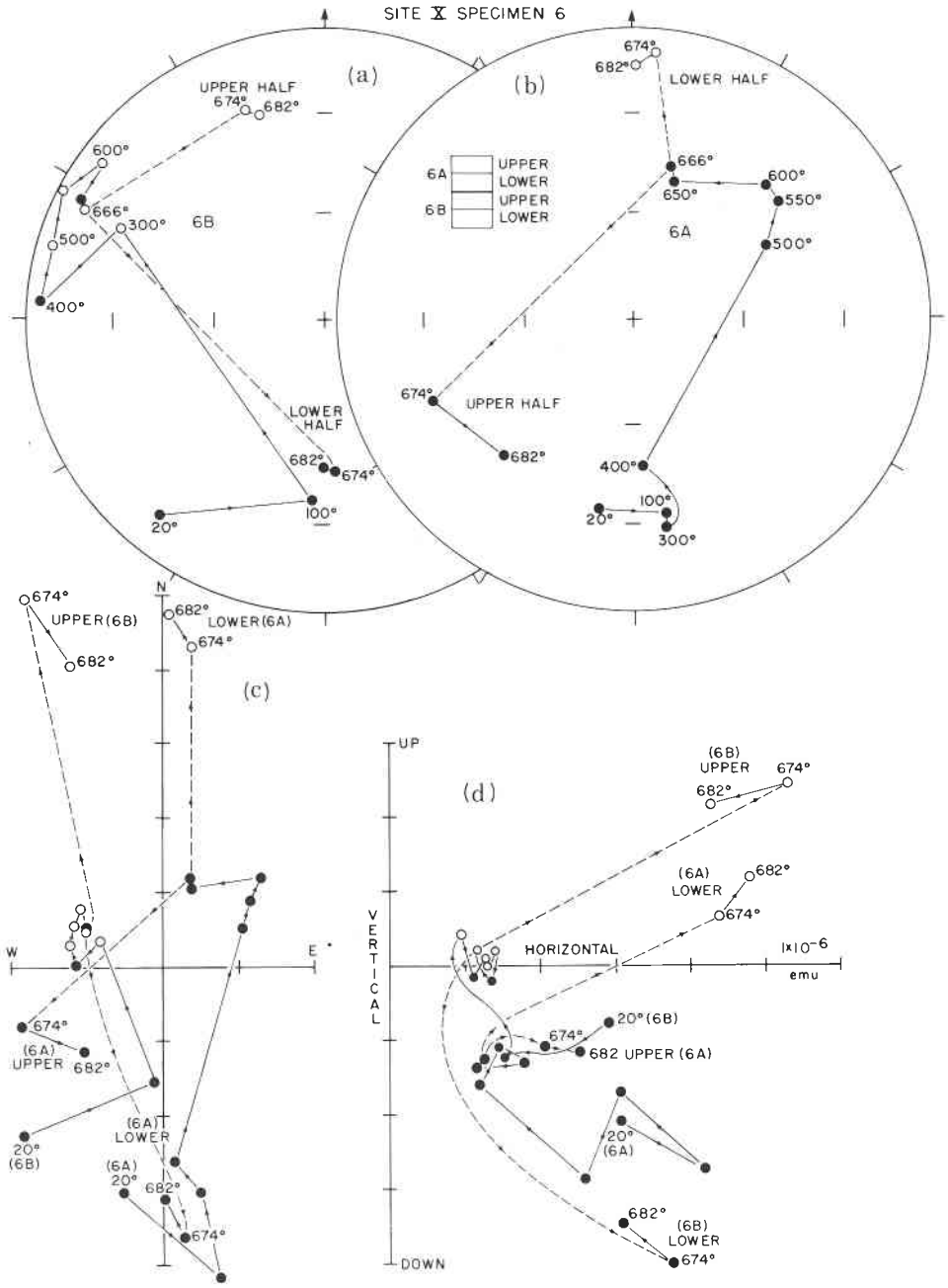


FIG. 9. Directions (a and b) and vector diagrams (c and d) of intensities of four different parts of a 4.4 cm (height) core after each thermal demagnetization step. The dashed lines join the two halves of specimens A and B which were cut after the 666 °C heating. The magnetization at each end of the core is southerly directed, and in the middle it is northerly directed. It is believed to be caused by two polarity reversals of a field, which originally directed towards the south, first reversed to the north and later reverted to its original polarity. This is supported by the fact that the intensities of the parts are larger than those of the whole specimens (d).

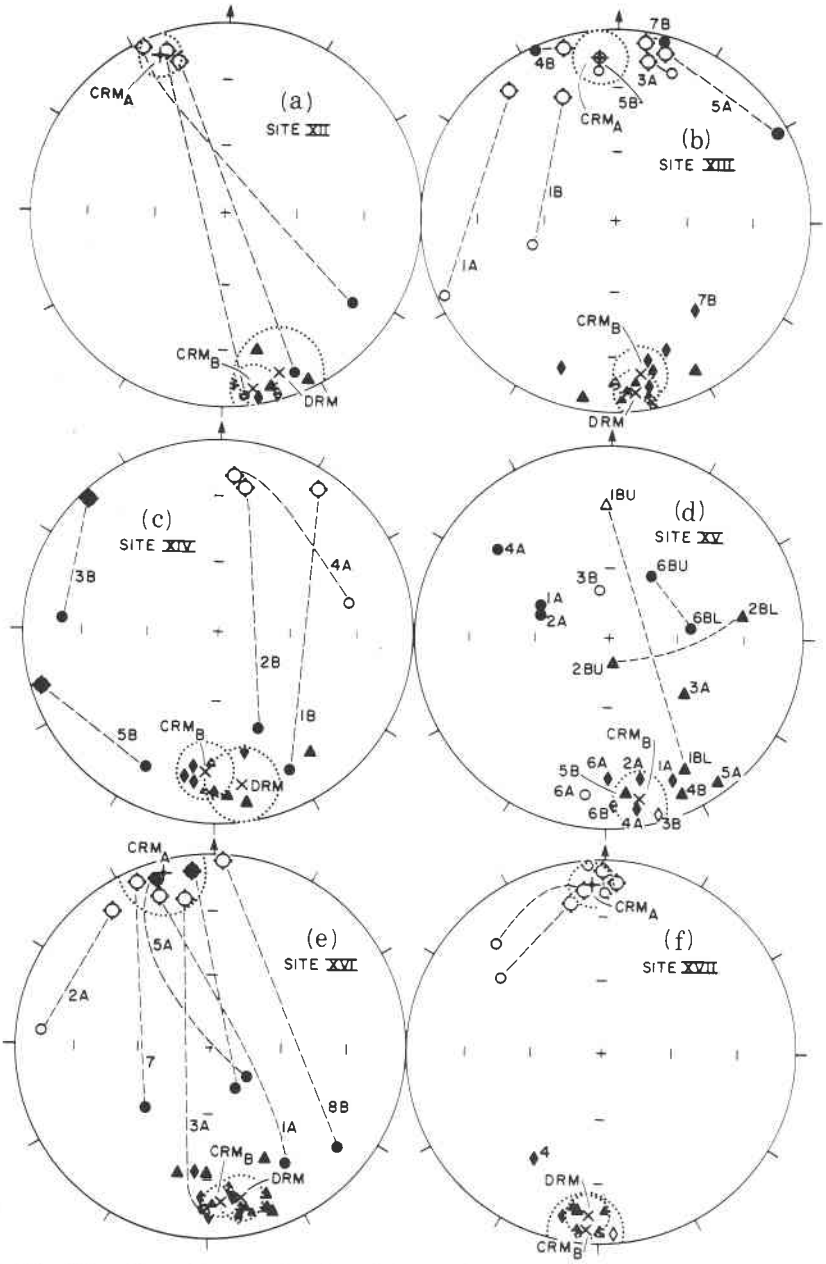


FIG. 10. Directions of magnetization of the last six sites sampled. Except for site XV, the dashed lines join the directions of RV: **DRM** + **CRM**$_A$ (small circles) measured after thermal demagnetization and the CRM$_A$ directions (large 4–wing circles) obtained by subtracting the DRM vector. It is evident that the grouping is much improved after the vector subtraction. The DRM, CRM$_A$, and CRM$_B$ specimen directions, the site mean directions, and the α_{95} circles of confidence are shown: At site XV, the dashed lines join the directions of magnetization of the halves of three specimens that were cut during treatment. Direction and intensity changes of some of these specimens are shown in more detail in following figures.

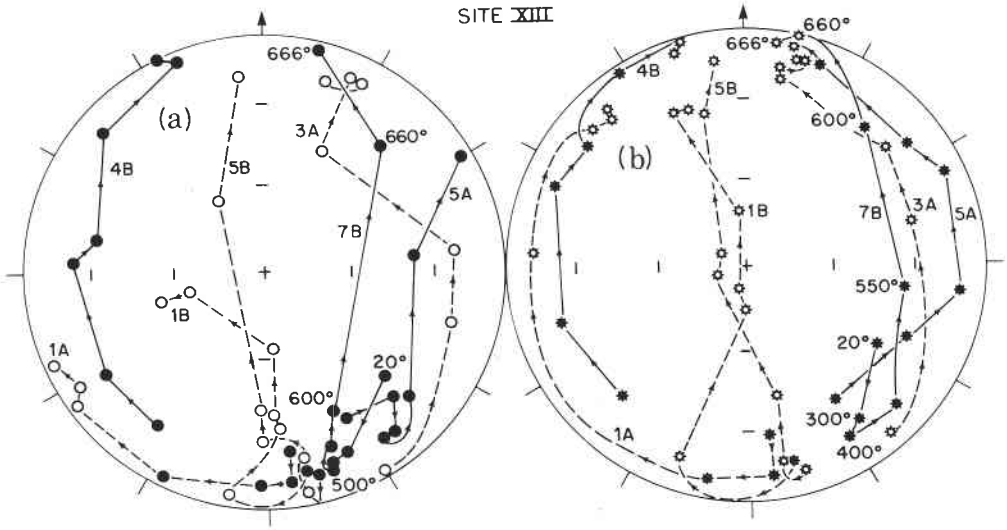


FIG. 11. Directions obtained after each demagnetization step for all specimens of this site are shown in (*a*). The directions in (*b*) are obtained by subtraction of the DRM vector from the vector whose direction is shown in (*a*). Except for 7B, the directions shown are for successive temperatures of 20, 400, 560, 600, 650, 660, and 674 °C. Part (*b*) shows that the CRM_A is definitely northerly directed and that the directions have reached an end-point. This evidence is lacking in (*a*) where the directions either keep changing considerably at high temperatures or reach an end-point in apparently random directions (*e.g.*, third quadrant for 1A and fourth for 4B). Clearly, the apparent inconsistency of the thermal results is not owing to the fact that the magnetization was not acquired along the magnetizing field, but rather to the fact that they represent the resultant vector of two opposite magnetizations that cannot be separated thermally. *Note*: The CRM_B direction of 7B was excluded from the calculation of the CRM_B site-mean direction (Table 4). The results show that the CRM_B of that specimen was acquired partly (low blocking temperatures) in a northerly directed field, partly (higher blocking temperatures) in a southerly directed field; thus indicating that the 20 °C direction is not an indicator of the field direction.

specimens; this excursion being usually 20 to 40° and, in a few instances, as much as 70 to 90°.

The usefulness of vector analysis is well illustrated in this group of sites. Without it, the directions obtained from thermal experiments alone (Fig. 10, circles) are quite scattered and appear for the most part meaningless.

At sites XIII, XVI, and XVII, the respective k values are 1.9, 2.6, and 7.6 (Table 4). The k of site XII is slightly better (11), but then the mean direction is southerly. However, when the southerly directed DRM is subtracted, the k values of these four sites increase by a factor of 8 or more (Table 4), thus showing that the CRM_A is definitely northerly directed. Some examples of the recovering of the CRM_A by vectorially removing the DRM are shown graphically (Figs. 11 to 13). At site XIII, the results obtained directly from the thermal experiments (Fig. 11*a*) are apparently useless for field determination purposes. Either the directions do not reach an end-point (5A, 5B, and 7B) or reach end-points in quite different directions: 3A in the first quadrant, 1A and 1B in the third quadrant, and 4B in the fourth quadrant. However, when the contribution of the DRM vector is removed (Fig. 11*b*), a definite end point is revealed for the CRM_A vector, and the overall precision (k) improves from 1.9 to 15. Although a display of directions on a stereogram is easily read, it does not show how the grouping of such highly scattered directions (Fig. 11*a*) can improve so much when the DRM vector is subtracted. The reason for this is that the intensities of the oppositely directed DRM and CRM_A at these sites are comparable (Table 4), so that the direction of RV: **DRM** + **CRM_A** may lie almost anywhere on a stereographic net unless

TABLE 4. Vector analysis of DRM, CRM_A, and CRM_B of the specimens of sites XII–XVII

Specimen	Treatment	NRM Mi	NRM D°,I°	RV: DRM+CRM_A THERMAL Mi	RV: DRM+CRM_A THERMAL D°,I°	CRM_A THERMAL Mi	CRM_A THERMAL D°,I°	CRM_B Mi	CRM_B D°,I°	DRM Mi	DRM D°,I°
					SITE XII						
1A	ch	2.08	156,+03							0.84	163,+09
1B	th	1.70	165,+03	0.30	171,+06	0.55	339,−11	1.40	164,+02		
2	ch	1.83	171,+16							1.09	165,+28
3	th	3.21	169,+06	0.18	155,+11	0.73	341,−15	3.04	170,+06		
4A	th	2.32	174,+13	0.04	124,+21	0.79	333,−04	2.29	175,+13		
4B	ch	2.89	178,+04							0.82	152,+05
MEAN		*2.34*	*169,+08*	*0.17*	*151,+13*	*0.69*	*338,−10*	*2.24*	*170,+07*	*0.91*	*160,+14*
(*k*)			(73)		(11)		(137)		(108)		(33)
					SITE XIII						
1A	th	3.86	173,+24	0.51	246,−04	0.93	321,−13	3.77	165,+25		
1B	th	4.18	169,+16	0.43	253,−53	1.04	337,−30	4.26	165,+21		
2A	ch	4.57	161,+19							0.84	176,+08
2B	ch	4.79	158,+29							0.96	171,+16
3A	th	3.28	150,−02	2.09	019,−17	3.00	011,−16	4.86	168,+06		
3B	ch	5.04	147,+27							0.97	175,+12
4A	ch	4.17	170,+27							0.85	179,+16
4B	th	3.91	214,+24	1.33	334,+01	2.11	343,−06	4.66	199,+20		
5A	th	5.66	153,+29	0.44	060,+05	1.17	013,−07	5.67	158,+28		
5B	th	3.28	167,+12	0.64	353,−19	1.57	353,−14	3.91	168,+13		
6A	ch	6.05	171,+27							1.00	168,+04
6B	ch	5.80	146,+25							1.03	151,+12
7A	ch	6.25	130,+43							0.84	189,+07
7B	th	5.05	133,+34	0.09	014,+05	0.93	009,−06	5.08	134,+34†		
MEAN		*4.71*	*160,+25*	*0.79*	*350,−19*	*1.55*	*353,−14*	*4.52*	*170,+19*	*0.93*	*173,+11*
(*k*)			(15)		(2)		(15)		(27)		(43)
					SITE XIV						
1A	ch	3.95	214,+24							1.25	175,+18
1B	th	3.55	182,+24	0.91	152,+20	0.53	035,−08	2.78	192,+24		
2A	th	2.99	166,+37	0.39	157,+44	1.29	009,−25	2.61	167,+36		
2B	ch	4.13	195,+28							1.63	183,+30
3A	ch	2.18	169,+27							1.49	183,+18
3B	th	4.28	215,+32	1.71	274,+20	2.16	315,+03	3.54	189,+28		
4A	th	2.91	176,+14	0.35	078,−32	1.08	006,−20	3.01	182,+17		
4B	ch	3.09	175,+30							0.98	169,+11
5A	ch	3.02	198,−02							0.68	142,+22
5B	th	4.12	192,+22	0.89	207,+20	0.81	252,+04	3.26	188,+22		
MEAN		*3.42*	*188,+25*	*0.85*	*175,+30*	*1.17*	*344,−14*	*3.04*	*184,+26*	*1.21*	*170,+20*
(*k*)			(18)		(1)		(2)		(52)		(23)
					SITE XV						
1A	th	4.76	157,+27	0.75	293,+59			4.77	154,+19		
1B	ch	4.01	155,+11							0.30	001,−31U
										1.02	149,+20L
2A	th	4.49	168,+28	0.36	289,+60			4.44	166,+24		
2B	ch	5.10	160,+20							0.97	168,+68U
										0.95	081,+30L
3A	ch	3.50	161,+08							0.74	126,+48
3B	th	3.20	163,−07	0.28	340,−68			3.28	163,−02		
4A	th	5.28	170,+13	0.36	308,+27			5.48	168,+11		
4B	ch	6.20	175,+12							0.97	154,+10
5A	ch	5.74	178,+13							0.55	142,+04

TABLE 4 (concluded)

Specimen	Treatment	NRM Mi	NRM D°,I°	RV: DRM+CRM$_A$ THERMAL Mi	RV: DRM+CRM$_A$ THERMAL D°,I°	CRM$_A$ THERMAL Mi	CRM$_A$ THERMAL D°,I°	CRM$_B$ Mi	CRM$_B$ D°,I°	DRM Mi	DRM D°,I°
					SITE XV (cont.)						
5B	ch	7.03	167,+09							0.88	171,+18
6A	th	4.43	178,+26	0.18	186,−18			4.30	178,+27		
6B	th	4.24	175,+16	0.78	034,+58U			4.29	177,+12		
				0.16	084,+54L						
MEAN		*4.91*	*167,+15*	*0.38*	*295,+47*			*4.44*	*168,+15*	—	—
(*k*)			(42)		(1)				(36)		
					SITE XVI						
1A	th	4.11	168,+16	0.09	148,+27	0.77	341,−17	4.03	168,+16		
1B	ch	4.73	165,+15							0.86	160,+18
2A	th	4.60	188,+16	0.36	274,−13	0.94	324,−15	4.61	183,+17		
2B	ch	4.20	176,+00							0.75	166,+12
3A	th	4.81	180,+11	0.63	182,+15	0.21	350,−23	4.18	180,+10		
3B	ch	3.64	172,+27							0.84	179,+17
4A	ch	3.79	282,+70							0.67	194,+32
4B	ch	4.08	175,+20							0.80	183,+32
5A	th	2.81	184,+35	0.27	135,+70	0.84	343,+06	2.62	186,+31		
5B	ch	4.51	175,+18							0.93	160,+10
6A	ch	3.60	163,+22							0.24	154,+35
6B	ch	3.68	171,+16							1.20	167,+14
7	th	4.20	176,+27	0.26	226,+51	0.66	337,−06	4.02	174,+25		
8A	ch	4.31	154,+12							0.70	160,+09
8B	th	4.31	168,+14	0.35	129,+16	0.44	003,−02	4.04	171,+14		
9	th	3.88	170,+26	0.38	145,+69	0.62	354,+08	3.62	171,+22		
MEAN		*4.08*	*174,+22*	*0.33*	*176,+47*	*0.64*	*344,−07*	*3.87*	*176,+19*	*0.78*	*169,+20*
(*k*)			(14)		(3)		(22)		(71)		(26)
					SITE XVII						
1A	ch	4.52	184,+05							2.61	180,+03
1B	th	5.41	174,−05	0.59	001,−16	3.19	000,−05	5.96	175,−03		
2A	th	7.32	189,+02	0.76	305,−35	3.13	347,−22	7.64	185,+05		
2B	ch	6.43	182,+14							2.59	177,+16
3A	th	2.53	197,+15	0.70	356,−02	2.55	004,−12	3.18	192,+12		
3B	ch	3.91	175,+03							1.87	187,+16
4	th	5.87	216,+33	0.62	315,−23	2.84	354,−17	6.10	210,+34†		
5	ch	8.94	176,+13							2.44	188,+19
MEAN		*5.62*	*186,+10*	*0.67*	*336,−21*	*2.93*	*356,−14*	*5.56*	*184,+05*	*2.38*	*183,+14*
(*k*)			(21)		(8)		(64)		(51)		(85)

NOTES: Code as per Table 2. U and L for upper and lower half of cut specimens.
†not included in mean, see captions of Figs. 11 and 13 for explanation.

the two directions are *exactly* anti-parallel which will be very rare if only because of secular variation. Because of this deficiency in stereo nets, the results are best illustrated by vector diagrams (Figs. 12 and 13). At these four sites (XII, XIII, XVI, and XVII), the CRM$_B$ was acquired in a southerly directed field (Fig. 10).

At site XV the CRM$_B$ was acquired in a southerly directed field, while both the DRM and the CRM$_A$ appear to have been magnetized during a field reversal (Fig. 10*d*). The results are similar to those of sites VIII to XI, and the slicing of specimens (sp. 1) shows that opposite magnetizations may reside within a single specimen.

At site XIV both the DRM and the CRM$_B$ were acquired in a southerly directed field. The CRM$_A$ is mainly northerly directed, but the directions are scattered, indicating that the magnetization probably occurred near the beginning or the end of a field reversal.

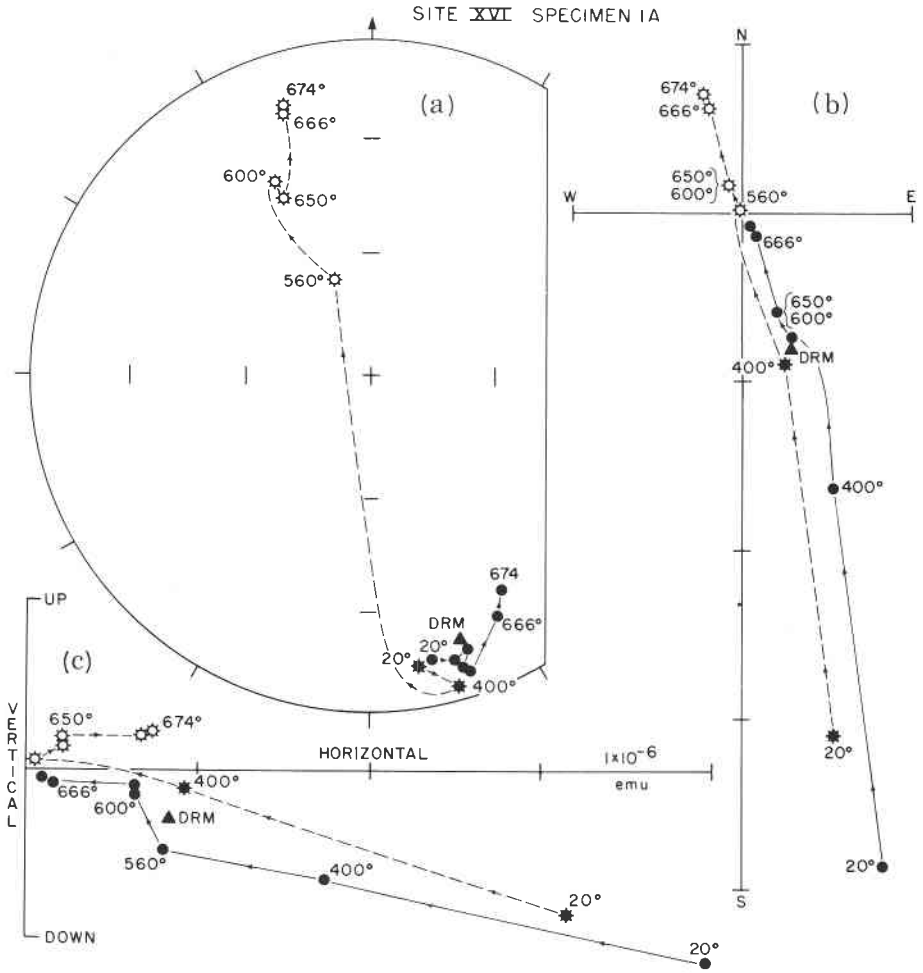


FIG. 12. Thermal minus DRM (8–wing circles) curve obtained by subtracting the DRM vector (triangle) from the vectors obtained thermally (circles). Diagrams (*b*) and (*c*) show that the presence of the northerly directed CRM_A was not evident in the thermal curve (circles) because the intensity of RV: **DRM** + **CRM_A** at high temperatures was almost nil owing to the fact that, as shown in (*c*), the DRM and the CRM_A (674 °C) are of about equal intensities. This figure also illustrates the deficiency of stereograms; it is difficult to visualize from (*a*) alone how the large direction change (≈120°) between the 400 and 560 °C directions of the thermal minus DRM curve (8–wing circles) could be derived from the small difference shown in the thermal curve (circles). The vector diagrams, however, clearly show that the horizontal component passes from the second to the fourth quadrant (*b*) and that the inclination changes from positive to negative (*c*) while in the thermal curve the direction remains in the second quadrant and positive with a continually decreasing intensity.

The exact number of reversals that occurred during the magnetization of sites XII to XVII cannot be established. The CRM_A of each site may represent six different reversals. Alternatively, it may be that more than one site was magnetized during the same reversal. However, it is noted (Fig. 10) that the circles of confidence are small at sites XII and XVII and larger at the sites inbetween. Likewise, the CRM_A precision (*k*, Table 5) decreases from sites XII to XV and thereafter increases. So, it seems that the north polarity of the CRM_A was interrupted at least once resulting in the more southerly directions at site XV. It is then suggested that at least two and possibly as many as six reversals are recorded in these sites.

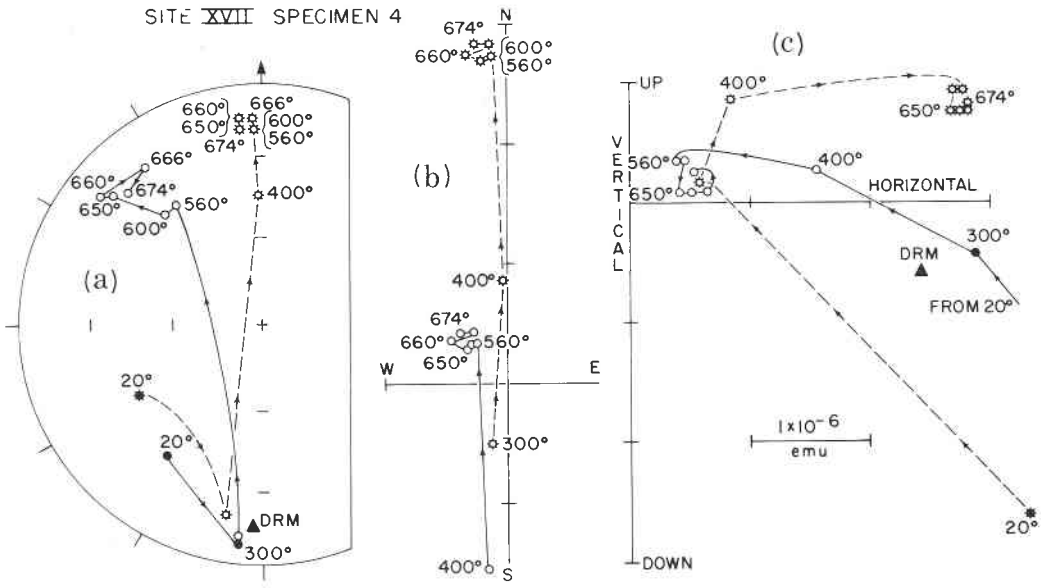


FIG. 13. Stereogram (*a*) shows that the dispersion of directions at high temperatures is much smaller in the thermal minus DRM curve (8–wing circles) than in the thermal curve (circles). Diagrams (*b*) and (*c*) show that this is because the intensity of the CRM_A is much larger than that of RV: **DRM** + **CRM_A**. Since the 8–wing circles are much further away from the origin than the circles, their angular dispersion is smaller. The directions in the low-temperature part of the curve indicate that the part of the CRM_B with low blocking temperatures ($\leqslant$300 °C) was acquired in a northerly directed field while the part with higher blocking temperatures was acquired in a southerly directed field. The 20 °C direction was then excluded for the calculation of the CRM_B mean-site direction (Table 4).

Viscous magnetization experiments

As mentioned previously, the chemical leaching experiments were performed in the earth's field where small viscous components can be acquired during treatment (Fig. 14; specimen 1B). This could be responsible for some of the differences between the paired thermally and chemically treated specimens observed at different stages (*e.g.*, at the transition point during the leaching of the chemical specimens); that is, *except for the last result.* Those viscous components decay rapidly (24–300 hr), when the specimen is placed in zero-field for 500 hr.

Experiments performed with thermally treated specimens led to a rather surprising result. Fifty specimens (demagnetized at 674 °C) were divided into two groups with thirty of them being subjected to chemical leaching in 10N HCl. They were measured at different intervals and replaced in an upright or inverted position in the earth's field as described in Fig. 14. Whether the rocks were subjected to chemical leaching (sp. 2A and 7A) or not (sp. 3), their susceptibility to the acquisition of viscous components appears to be the same, and, most importantly, when these viscous components are removed (by decay in zero-field) the original 674 °C result (both direction and intensity) is recovered. So it seems that the thermal treatment renders the grains carrying the CRM_A insoluble in HCl. This may be owing to annealing (Dunlop 1971).

TABLE 5. Precision *k* of DRM, CRM_A and CRM_B of sites XII–XVII

SITE	DRM	CRM_A	CRM_B
XII	33	137	108
XII	43	15	27
XIII	23	2	52
XIV	—	—	36
XVI	26	22	71
XVII	85	64	51

The experiments also show that (*1*) the viscous components are carried mostly by the CRM_A-carrying grains, (*2*) the DRM-carrying

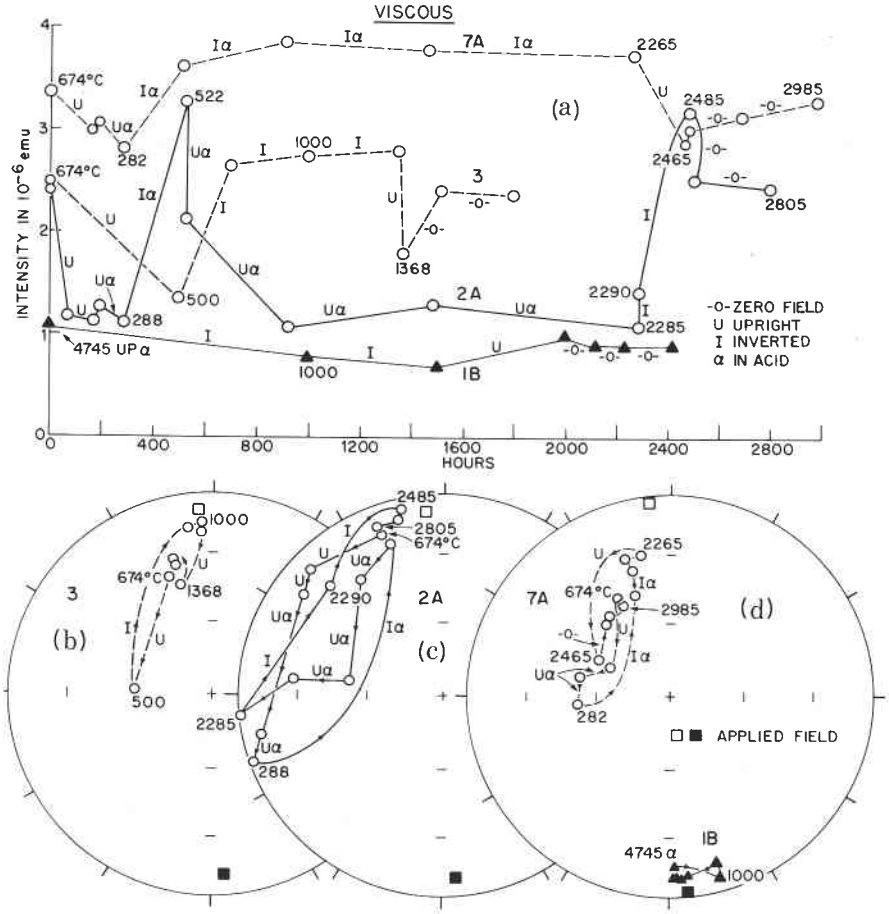


FIG. 14. Acquisition and removal of viscous magnetization. The specimens were left in an upright or inverted position for different periods of time (as indicated) in the earth's field (0.6 Oe). As in the previous figures, the directions are shown after correction for the attitude of the specimens in the field and correction for bedding. The direction of the earth's field (applied field) has been corrected accordingly. Three (2A, 3 and 7A) of the specimens were previously demagnetized at 674 °C and the results obtained begin from this point (0 hr) in this figure; 2A and 7A were immersed in 10N HCl for 2285 and 2265 hr respectively as indicated by (*a*). The last few readings were obtained after leaving the specimens in a zero-field for a few hundred hr. For these three specimens, the first and last readings are identical (within the limits of error) in both direction and intensity. This shows (*1*) that viscous magnetizations are effectively removed by leaving the specimens in zero-field, and (*2*) that the 674 °C thermal treatment has frozen or altered the CRM_A-carrying grains so that the CRM_A cannot be removed by chemical leaching. Specimen 1B shows that viscous components acquired during chemical leaching decay rapidly in a zero-field. The original reading (0 hr) was obtained after leaching the specimen for 4745 hr in the earth's field in an upright position. The last three readings obtained after leaving the specimen in zero-field for 120, 240, and 420 hr are nearly identical, thus indicating that the viscous magnetization was removed in less than 120 hr.

grains can acquire only small viscous components over a long leaching period (sp. 1B), and (*3*) the CRM_B-carrying grains appear incapable of acquiring substantial (if any) viscous magnetizations.

The last point (*3*) is illustrated as follows. When the magnetization of the fifty specimens is investigated, there is no clear indication that the two groups (leached and unleached) can be magnetically differentiated. The magnitude and the rate of acquisition of the viscous magnetization, and the ratio of viscous to stable magnetization vary amongst specimens but independently of the group to which the speci-

men belongs. For example (Fig. 14), the leached specimens 2A and 7A are respectively more, and less, receptive to viscous magnetization than the unleached specimen 3. The physical appearance of specimens from each group is, of course, different; the leached specimens having lost their redness thus suggesting that the CRM_B-carrying grains have been destroyed; and the unleached specimens having retained their redness, thus indicating that the CRM_B-carrying grains, although demagnetized, are still present unless they have been chemically altered by heating. The fact that the two groups, one with and one without the red cement, are magnetically comparable, indicates that the CRM_B-carrying grains (or their altered version) are not appreciably affected viscously.

Polished sections

Specimens which had been (*a*) chemically leached for more than 3000 hr, (*b*) thermally demagnetized at 674 °C, and (*c*) partially leached (250–410 hr) (whose directions were near the transition point) were examined in reflected light. The (*a*) specimens show no trace of red material and it appears that the only possible carriers of magnetization left in the specimens are small (0.2–6μ) particles—some rounded but most of them elongated—of specular hematite which compose 0.02 to 0.1% of the rock. It is believed that these unleachable particles were the first magnetic carriers to appear during formation of the rock to form a DRM.[4] The (*b*) specimens consist of grains of different sizes (up to 60μ) imbedded into the red cement; small hematite particles being disseminated in both the red cement and in most grains. The amount of hematite is much larger (in both number and size of particles) than in the (*a*) specimens, and composes at least a few percent of the rock. The (*c*) specimens can be qualitatively described as similar to the (*b*) specimens except that the red coloring has disappeared. The CRM_B is then associated with the red cement consisting of extremely small particles beyond the resolving power of the microscope. The CRM_A carrier is believed to be in the excess amount of hematite in the (*b*) and (*c*) as compared to the (*a*) specimens. This carrier may either reside in the smaller particles, which finally disappear upon further leaching, or in an accretion around the detrital particles which, although less soluble in acid than the red cement, is eventually dissolved. The fact that the detrital particles appear insoluble (the DRM remaining constant for the last 2000 to 3000 hr) favours the latter.

Synthesis and Conclusions

The vector analysis of results obtained from thermal demagnetization and chemical leaching allows us to evaluate the reliability of the results obtainable from those two cleaning techniques when used separately or together, with or without vector analysis.

Thermal experiments

Partly because thermal experiments have become a standard procedure in the study of magnetization of red beds, an important aspect of this present study has been the thermal stabilities of the DRM and of the CRM_A. Neither can be completely demagnetized by heating near the Curie point of hematite and therefore neither can be thermally separated. The results show that the vector obtained after demagnetization at 674 °C is the resultant vector of (at least part of) the two magnetizations, so that its direction is not the field direction at the time of magnetization.[5]

[4]In this article, 'DRM' is taken to include both depositional and postdepositional DRM (Irving and Major 1964). The blocking temperatures indicate that hematite is probably the magnetic carrier. This hematite may have been present in the source rock or may have been derived from the oxidation of magnetite during erosion, transportation, deposition or soon after deposition but prior to partial solidification of the medium; that is, as long as the magnetic particles were able to rotate into the direction of the ambient field. In this context, the remanent magnetization (described as DRM) is due to a mechanical process independently of the origin of magnetization of the constituents. It is noted that if the remanence of the constituents is of chemical origin (*e.g.*, see Collinson 1965*a*, *b*; Van Houten 1973), their solubility spectrum is clearly different from the solubility spectra of the CRM_A and CRM_B. Therefore chemical leaching provides a means of distinguishing the three magnetizations.

[5]The direction of the resultant vector would be parallel to the direction of the field if the DRM and the CRM_A were exactly parallel or anti-parallel. Since the DRM and the CRM_A are acquired at different times, this means that either the field direction remained constant (no secular variation during that time) or that, during the acquisition of the CRM_A, the field resumed the direction it had during the acquisition of the DRM. Although this may happen at the odd site, it is most improbable that it will happen on a regular basis at all sites.

When the DRM and the CRM_A are oppositely directed, the angular difference between the directions of the (measured) resultant vector and of the field may vary from small to very large. At sites I, IV, V, and VI the angular differences between RV: **DRM** + **CRM_A** (given by the thermal results) and the CRM_A (Table 2) are 8°, 5°, 5°, and 6° respectively. The difference is small owing to the fact that the CRM_A is much larger ($\simeq$ 4 times) than the (not yet uncovered) DRM so that the biasing effect of the DRM is small. At sites XII to XVII (XV excepted) the angular differences between RV: **DRM** + **CRM_A** and the CRM_A at the specimen level are extremely large (Fig. 10). This is owing to the fact that the DRM and the CRM_A have comparable intensities. The differences between the site mean directions (Table 4, between RV: **DRM** + **CRM_A** and CRM_A) are also large; the differences being so large at some sites (XII, XIV, and XVI) as to show the wrong polarity. The RV: **DRM** + **CRM_A** precision (k) for those sites varies from 1 to 11 and clearly, in this instance, the directions obtained solely from the thermal experiments do not provide any reliable estimate of the direction of the magnetizing field.

On the whole then it appears that, when the polarities of the DRM and the CRM_A are opposite, the value of thermal results for field determination is dependent upon the intensity ratio of CRM_A/DRM. When this ratio is near unity, the results are meaningless (sites XII to XVII). As the ratio increases, a determination with a reasonable degree of accuracy may be obtained as shown by the average of the site-mean directions (sites I, IV, V, and VI) of the thermal results (RV: **DRM** + **CRM_A**) which differs from that of the CRM_A by only 8°.

Even when the DRM and the CRM_A have apparently the same polarity, the direction obtained from the thermal experiments may differ substantially from the direction of the field. For example, the RV: **DRM** + **CRM_A** of site II is 18° away from the field direction as given by the DRM and CRM_B results. This difference is probably attributable to the effect of a field reversal occurring in the early or late stage of the CRM_A acquisition.

Results similar to these thermally obtained results are often found in the paleomagnetic literature (for references, see Roy and Robertson 1968; Roy and Park 1972). It appears then that the lengthy process of magnetization described here is not a pecularity of the Hopewell red beds and may be quite common. It is then concluded that results obtained from thermal studies of red beds magnetized at the time of field reversals may not be sufficient to obtain an accurate determination of the field direction.

It is pointed out, however, that with the aid of vector analysis, the CRM_B can be obtained by subtracting the thermal results from the NRM results. The high precision (k) of the CRM_B results (Table 6) indicates that the mean direction is most probably indicative of the field direction at the time of magnetization. If this time can be established within reasonable limits (as in this instance), the CRM_B can then be used to determine (assuming a dipole field) the pole position at that time. In paleomagnetic work, the direction obtained after thermal demagnetization is usually considered as the most important and the NRM vector is often overlooked. The present results show that, in this instance, the NRM vector minus the vector remaining after demagnetization at high temperature yields a more accurate field direction than the direction of the vector remaining after thermal demagnetization.

Chemical experiments

Chemical leaching can isolate the DRM, and therefore the pole position at the time of deposition can be determined. In addition, when the transition point between oppositely directed CRM_A and CRM_B can be established, the direction of both can be determined by vector analysis. However, depending on the leaching rate, the transition point may sometimes be easily missed (sites XII–XVII). Furthermore, when the CRM_A and the CRM_B have the same polarity (sites II, III, and VII), the transition point cannot be obtained.

Combining chemical and thermal experiments

Since each of the cleaning techniques is more efficient for uncovering a different part of the record contained in these rocks, it is evident that the two treatments complement each other. Although chemical experiments may be sufficient to determine a pole position, thermal experiments remain of great value. The end-point usually obtained from thermal experiments is generally a more accurate indicator of the direction of RV: **DRM** + **CRM_A** than

TABLE 6. Compilation and averages of the site-mean directions of the DRM, CRM_B, and CRM_A (whose $k > 15$)[+] given in Tables 2 and 4[x]

SITE	DRM D°,I°	θ_{63}	CRM_B D°,I°	θ_{63}	CRM_A (NORTH) THERMAL D°,I°	θ_{63}	CRM_A (NORTH) CHEMICAL D°,I°	θ_{63}	CRM_A (SOUTH) D°,I°	θ_{63}
I	179,+06	5.6	175,+07	8.6						
II	170,+17	10.4	178,+11	6.2						
III	175,+10	12.5	173,+14	8.1					188,+14	17.1
IV	169,+05	8.8	171,+11	12.1	352,−22	10.2	006,−17	9.8		
V	179,+15	12.3	180,+21	6.7	345,−02	19.6	348,−06	11.1		
VI	180,+21	5.9	167,+27	5.9	351,−28	5.6	337,−25	12.4		
VII	174,+14	9.7	177,+22	6.6					170,+13	7.1
XII	160,+14	14.0	170,+07	7.8	338,−10	6.9				
XIII	173,+11	12.3	170,+19	15.7	353,−14	20.8				
XIV	170,+20	17.1	184,+26	11.1						
XV			168,+15	13.5						
XVI	169,+20	15.7	176,+19	9.6	344,−07	17.1				
XVII	183,+14	8.8	184,+05	11.3	356,−14	10.1				
AVERAGE	173,+14	8.1	175,+16	9.0	348,−14	10.7	351,−16	16.8	179,+14	12.4
R	11.891		12.850		6.895		2.914		1.923	
k	101.		80.		57.		23.		—	
α_{95}	4.3		4.7		8.1		26.2		—	

NOTES:
[+]The CRM_A of sites I and II have been excluded because it appears to have been acquired near the beginning or the end of a field reversal and is probably composed of the oppositely directed components (see text). If so, the direction of the resultant vector is not parallel to the direction of the magnetizing field.
[x]The few site-mean directions given in Table 3 have small k and are not considered to represent the direction of the field at the time of magnetization. R = resultant vector (Fisher 1953); α_{95} radius of the circle of confidence P = 0.05.
θ_{63} = Circular standard deviation.

is the difficult-to-determine transition point of chemical experiments.

In summary, the study indicates that it is preferable to use thermal results for CRM_B determination, chemical results for DRM determination and thermal (RV: **DRM** + **CRM_A**) minus chemical (DRM) results for CRM_A determination. In certain instances (sites I, IV, V, and VI), the directions of the CRM_A and CRM_B can also be determined from the chemical experiments. This then provides a check on the thermal results and increases the accuracy of the determination of those two directions.

It is shown, in the following discussion, that vector analysis of the combined chemical and thermal results yields not only a more accurate field direction determination but also much information about the process of magnetization of red beds and about certain features of the earth's field. Much additional information about field reversals can be obtained by studying the within-specimen magnetization.

Field direction indicators

At most sites reliable indicators of the field direction could be obtained, and these are listed in Table 6 with the relevant statistics. Since the DRM has been isolated, its direction should be considered as the most reliable. However, because the CRM_B is more strongly magnetized resulting in relatively smaller measurement errors, and since the viscous components are negligible, the CRM_B direction is considered equally reliable. The direction of the CRM_A is considered less accurate owing to inaccuracies in the determination of the vector length. However, the near anti-parallelism of the directions clearly shows the reversal was completed through 180°.

Relative timing of DRM, CRM_A, and CRM_B

Since at several sites (Table 6) the polarity of the CRM_A is opposite to those of the DRM and CRM_B, it is evident that the CRM_A was not acquired at the time of magnetization of the DRM or CRM_B.

Evidence that the DRM and CRM_B were not acquired at the same time is found at sites X (Fig. 7*c*), XI (Fig. 7*d*), and XV (Fig. 10*d*), where the polarities of the DRM are found to be mixed while all the CRM_Bs are southerly directed. Comparison of the DRM and CRM_B precisions (k; Table 5) also indi-

TABLE 7. Composite site-mean directions calculated from Table 6 results

	N	D°,I°	R	κ	α_{95}	θ_{63}	POLE POSITION E°,	N°	δp	δm
CRM_A (North)	10	349.0,−14.6	9.805	46	7.2	11.9	129.1,	35.9	3.8	7.4
$DRM+CRM_B$	25	174.2,+14.9	24.737	91	3.1	8.5	122.7,	36.4	1.6	3.2
$DRM+CRM_B+CRM_A$ (South)	27	174.5,+14.8	26.707	89	3.0	8.6	122.3,	36.5	1.6	3.1

δp and δm are the semi-axes of the ellipses of confidence (P = 0.05).

cates different ages for the two magnetizations; for while the DRM k reaches a minimum at site XV, the CRM_B k remains high. It is then concluded that the three magnetizations were acquired at different times.

There are indications that, of the two CRM's, the CRM_A was acquired first. At sites X, XI, and XV, both DRM and CRM_A have northerly and intermediate directions, while the CRM_B is constantly directed towards the south; hence indicating that, of the two CRMs, the CRM_A is the more closely related in time to the DRM. The precisions (k) of the three magnetizations (Table 5) also show the closer relationship between the DRM and the CRM_A; both of them reach a minimum at site XV while no clear reduction of precision is found for the CRM_B. It is then suggested that the DRM, CRM_A, and CRM_B were acquired in that order. This means that when the CRM_A direction is opposite to those of the DRM and the CRM_B, as at sites I and IV to VI for example, the DRM was acquired before the (first) field reversal, the CRM_A after the (first) field reversal and the CRM_B after a second field reversal when the field has reverted to its original polarity.

Process of magnetization

The directions of the three magnetizations (Table 6) are nearly parallel (irrespective of sign); being near the Hopewell direction (178°, + 20°) obtained from sites from both sides of the Maringouin anticline (Roy and Robertson 1968; Roy and Park 1969). A fold test showed that the magnetization yielding that direction predated the folding that occurred within 35 m.y. of deposition. This sets an upper time limit for the acquisition of these three magnetizations. Although these magnetizations were acquired at different times, none of them can be considered as secondary; that is, in the sense often used to describe a magnetization acquired during some tectonic event as, for example, during deep burial. Instead, we propose that each of them is part of, and was acquired at different stages of, a continuing and lengthy process of magnetization which started at the time of deposition of these beds. The first stage was probably the alignment of small specular hematite particles (observed in the specimens leached for more than 3000 hr) along the field during or shortly after deposition to form a DRM. The next stage may have been the acquisition of the CRM_A by hematite growth around the deposited particles, and the final stage, the acquisition of the CRM_B during and/or after filling and cementing of all grains during final lithification.

A Carboniferous field direction

The DRM and CRM_B directions (Table 6) are almost identical, and it appears that the pole remained stationary during the whole magnetization process. Different combinations using site-mean directions are given in Table 7. The south direction rather than an average of north and south directions is considered a better estimate of the field direction during and shortly after deposition of these Mississippian beds (Bell 1944). The small difference (5°) between the anti-parallelism of north and south directions may be owing to secular variation, but is more probably caused by inaccuracy in the determination of the CRM_A vector.

The northerly directed CRM_A may have recorded the last negative polarity(ies) of the field that occurred before the long positive-polarity interval (Kiaman) as recorded by Upper Carboniferous–Permian rocks. Since no negative polarity has been found in the Cumberland Group (Table 1), this places the lower boundary of that (positive-polarity) interval between the Late Mississippian and the Middle Pennsylvanian.

Secular variation

The dispersion of site means is commonly

used to estimate the magnitude of secular variations in the past and to determine, by comparison with the dispersion of the present field, the relativeness of past and present secular variations (see Irving 1964, p. 146). The standard deviation of the southerly directed magnetizations (Table 6) is much smaller (by about half) than the deviation of 17° attributed to secular variation of the present field (Creer 1955) at that latitude. Although it cannot be ruled out, we do not believe that this shows that secular variations in Carboniferous time were twice as small as present secular variations. Our explanation is that, owing to the lengthy process of magnetization, the mean direction of a specimen is not a spot reading of the field but rather an average of some *portion* of the secular variation cycle. This integration of secular variations within specimens results in a reduction of both within-site and between-site dispersions and probably explain the small standard deviations of the DRM and CRM_B (Table 6). Therefore it appears that, when the magnetization is acquired in a single-polarity field, secular variations calculated from red beds will tend to be too small. This also means that the effect of secular variations is reduced and that the mean field direction is determined more acurately; this explanation thus accounting for the small $\alpha_{95} = 3°$ obtained for the southerly directed magnetizations (second and third entries of Table 7).

The standard deviations (and circles of confidence) for the northerly directed CRM_A are, in general, larger (Table 6). This is expected since the CRM_A vector of some of the specimens composing a site cannot be determined accurately thus increasing the within-site dispersion. Therefore those standard deviations of magnetization acquired near a field reversal cannot be attributed to secular variations solely.

Field reversals

The thermal and chemical experiments, the vector analysis, and the results obtained by slicing specimens all point to a single explanation for the magnetization of these red beds; the magnetization of all specimens being acquired at different stages in a field (directed along 174°, + 15°) which reversed through 180° repeatedly. This is clearly indicated by vector analysis at sites I–VII, XII–XIV, and XVI–XVII. At the other sites the additional slicing of specimens and the $\Delta\Psi$ index indicate that, although the mean (within-specimen) direction may differ considerably from the field direction (174°, + 15° or 354°, − 15°), the magnetization was probably acquired along that direction,—that is, these specimens were magnetized partly before and partly after a field reversal. The results provide much information about field reversals, some of which is applicable specifically to field reversals during the Carboniferous and some of which may be applicable to field reversals in general.

Near the Mississippian–Pennsylvanian boundary the field was mainly southerly directed, and many reversals took place in less than a 35 m.y. period. At least four reversals are required to account for the northerly directed CRM_A of sites I, IV–VI, VIII–XIV, and XVI–XVII. However they may have been more numerous. The CRM_A appears to have been acquired shortly after the DRM, and it is possible that the alternating south and north CRM_A from site to site within a group (sites IV–VI, for example) are recording a cyclic event with the CRM_As of different sites representing different reversals. If the CRM_A of each site is taken to represent different reversals, ten (recorded) or more (unrecorded) reversals could have occurred. A closer spacing of sites is required to obtain a better estimate of the number of field reversals that occurred between the times of magnetization of sites I–XVII.

There is much evidence to suggest that field reversals are rapid and sometimes of short duration. The fact that, of three magnetizations recorded at so many sites, none of the directions appear to have recorded a truly intermediately-directed field—that is, near the east–west axis or very steeply inclined—suggests a rapid transition of polarity. The only evidence of a magnetization (CRM_A) acquired during transition is found at site I. However, the rate of acquisition of the CRM_A is unknown and so the time taken for the field to reverse cannot be determined from that result. At sites VIII–XI and XV, the analysis of intensities, directions, and $\Delta\Psi$ before and after cutting specimens, indicates that a complete reversal may be recorded by either the CRM_A or the DRM

within a specimen. It appears then that, in some instances at least, the field might reverse in less time than that taken for the deposition of one or two cm of sediments. The same results, and more specifically sample 6 of site X, also indicate that field reversals may be short-lived. These features can be explained by supposing that the field sometimes oscillates between normal and reverse polarities before stabilizing in one direction.

The future of chemical work

The usefulness of chemical leaching for the study of these red beds is clearly demonstrated. Much of the information contained in these rocks, which could not be retrieved thermally, becomes available with chemical leaching.

Little chemical work has been done on red beds (for other studies, see Collinson 1966 and Burek 1971) and as yet no standard procedure has been developed. It is probable that different acid concentrations and different techniques may have to be used depending on the porosity of the rock. For example, at sites XII to XVII, the CRM_A could possibly have been obtained with a weaker acid concentration. In other instances, because of the low porosity of the rocks, chemical leaching by immersion may not be practical (Roy and Fahrig 1973) and a pressure type device such as used by Burek (1971) is probably more efficient.

In view of these results it appears that the sampling technique for red beds could be improved upon. The number of reversals could have been better estimated with closer spacing of sites. More details about the behaviour of the field during a reversal might be obtained by extending the stratigraphic height at a site or by obtaining a very long core. Furthermore, because the within-specimen uniformity of direction can often be related to field reversals, it is preferable, whenever possible, to core the specimen perpendicular to bedding.

Synopsis

The conclusions drawn from the study of the magnetization of these Mississippian red beds are summarized below.

About the magnetization of red beds

(*1*) The magnetization of red beds is a lengthy process.

(*2*) Up to three magnetizations can be produced during that process.

About field direction determination

(*3*) One of the magnetizations can be isolated by prolonged chemical leaching so that a field direction can be determined directly.

(*4*) None of the magnetizations can be *isolated thermally*, so that, in certain instances, a field direction cannot be obtained from the thermally cleaned results alone.

(*5*) Attention is drawn to the NRM results, since it may be possible to determine a field direction by vector analysis of the results obtained before and after thermal demagnetization.

About the information recorded in red beds

(*6*) Because of (*1*), red beds can provide much information about certain features of the earth's field in the past.

(*7*) However, because of (*1*) and (*2*), a vector analysis of combined chemical and thermal results is required to retrieve that information.

(*8*) Near the Mississippian–Pennsylvanian boundary and before Cumberland time the direction (174°, + 15°; irrespective of sign) of the field remained stationary (within the secular variation) for perhaps several million years.

(*9*) During that time several field reversals occurred.

(*10*) Those field reversals appear to have recorded the last negative field polarity that occurred before the long positive polarity interval of the Upper Carboniferous–Permian.

(*11*) Because of (*1*), it is doubtful that red beds can provide a reliable estimate of secular variation in the past.

(*12*) After vector analysis of the different magnetizations, there is no evidence that the intensity of the field decreases at the beginning of a field reversal, indicating that, if the field intensity does decrease during a reversal, the lowering of intensity is of much shorter duration than it appears from direct observation.

(*13*) Vector analysis of within-specimen magnetization shows that field reversals can be quite rapid since they can be recorded within a (2.2 cm height) specimen.

(*14*) Field reversals may sometimes be of short duration as shown by the occurrence of two field reversals (that is, a positive–negative

polarity sequence) during the period of the chemical magnetization of 4.4 cm of sediments. (*15*) From (*13*) and (*14*), it is suggested that at reversal time the field may be unstable and may be subjected to an oscillatory motion.

It is realized that (*1*) and (*2*) may be applicable only to the two formations studied. However, we believe that the process of magnetization proposed here is common in red beds; this conclusion being mainly based on the fact that the thermal stability of two (DRM and CRM_A) magnetizations acquired at different times (sometimes before and after a field reversal) provides a coherent explanation of otherwise inexplicable thermally cleaned data. If that process of magnetization is applicable to most red beds, then those rocks are ideally suited for the study of field reversals. Whether or not items (*1*) to (*5*) can be generalized to include most red beds can be established by vector analysis of thermal and chemical experiments of other formations.

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Appendix

All distances between sites and specimens are relative stratigraphic heights. The specimens are given in numerical order with the lower number being stratigraphically lower; the distance between two samples is given in brackets in cm.

Site I: 1(5) 2(45) 3(30) 4(30) 5(15) 6(30) 7 as shown in Fig. 4.

Site II: 29 m above Site I; 1(10) 2(10) 3(5) 4(16.5) 5(1) 6(5) 7(30) 8(15) 9(2.5) 10.

Site III: 88 m above Site II; 1(1) 2(1) 3(1) 4(0.5) 5(10) 6(2.5) 7(2.5) 8(2.5) 9.
Site IV: 207 m above Site III; 1(2.5) 2(5.0) 3(7.5) 4(7.5) 5(2.5) 6.
Site V: 60 m above Site IV; 1(1) 2(2.5) 3(7.5) 4(7.5) 5(2.5) 6(1) 7(4) 8(2.5) 9(7.5) 10(10) 11.
Site VI: 29 m above Site V; 1(2.5) 2(43.5) 3(2.5) 4(2.5) 5.
Site VII: 60 m above Site VI; 1(0.5) 2(0.5) 3(1) 4(1) 5(5) 6.
Site VIII: 15 m above Site VII; 1(2) 2(8) 3(12) 4(20) 5. The angle between the coring axis of the cut specimen 3 and the bedding plane is 80°.
Site IX; 0.9 m above Site VIII; all specimens from same level. The angles between the coring axis and the bedding plane for the cut specimens 1 and 3 are 70° and 56° respectively.
Site X: 0.6 m above Site IX; 1(7.5) 2(7.5) 3(7.5) 4(6.5) 5(1) 6. The angles between the coring axis and the bedding plans for the cut specimens 1, 3, 4, 5, and 6 are 61°, 59°, 55°, 57°, and 57° respectively.
Site XI: 3 m above Site X; 1(1) 2(5) 3(10) 4(5) 5. The angle between the coring axis of the cut specimen 3 and the bedding plane is 60°.
Site XII: 6.1 m above Site XI; 1(10) 2(60) 3(10) 4.
Site XIII: 6.1 m above Site XII; 1(5) 2(5) 3(5) 4(5) 5(10) 6(10) 7.
Site XIV: 6.1 m above Site XIII; 1(7) 2(7) 3(7) 4(7) 5.
Site XV: 3 m above Site XIV; 1(10) 2(30) 3(47) 4(30) 5(40) 6. The angles between the coring axis and the bedding plane for the cut specimens 1, 2, and 6 are 72°, 61°, and 58° respectively.
Site XVI: 15 m above Site XV; 1(7) 2(3.3) 3(1.5) 4(8) 5(5) 6(3.5) 7(1.5) 8(8) 9.
Site XVII: 20 m above Site XVI; 1(15) 2(120) 3(30) 4(15) 5.